

CLASSICAL MECHANICS

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This notes is mostly about Lagrangian formulation and an introduction to Hamiltonian formulation of classical mechanics. Compared to Newtonian formulation, Lagrangian and Hamiltonian formulations are often easier, more powerful, and are better connected to other fields of physics (statistical, quantum, general relativity, etc.).

1 CALCULUS OF VARIATIONS

1.1 An Example

Suppose there are two points (x_a, y_a) and (x_b, y_b) on a two dimensional plane. What is the shortest path between the two points? The answer is obviously a straight line. This is usually taken as an axiom in geometry, but it can be proved using calculus of variations. More generally, if the two points are not on a plane, but on a curved surface, the shortest path (a [geodesic](#)) is no longer trivially a straight line.

We can formulate the problem in another way: let the shortest path be the function $y(x)$. What is the function y that minimizes the path length? It is known that the path length along a curve is given by

$$L[y] = \int_a^b \sqrt{dx^2 + dy^2} = \int_a^b \sqrt{1 + y'(x)^2} dx. \quad (1.1)$$

One of the classical calculus problems is to find the minimum point of a function. We do this by saying any infinitesimal change in y would have no effect on it, i.e. $dy/dx = 0$. Same work for the geodesic problem. Consider an infinitesimal deviation, or [variation](#), $\delta y = \epsilon \eta(x)$ of the function $y_0(x)$:

$$y = y_0 + \delta y = y_0(x) + \epsilon \eta(x).$$

We want to find a function $y_0(x)$ such that $\epsilon \eta(x)$ would have no effect on y_0 . Here ϵ is an infinitesimal dimensionless number, and $\eta(x)$ is an arbitrary function (not necessarily small). Intuitively, ϵ controls the size of variation, while $\eta(x)$ controls the shape. The only constraint is that $\eta(x_a) = \eta(x_b) = 0$ because the end points are fixed. Now substitute this y into (1.1) and Taylor expand in ϵ ,

$$\begin{aligned} L + \delta L &= L[y_0 + \epsilon \eta] = \int_{x_a}^{x_b} \sqrt{1 + [y_0'(x) + \epsilon \eta'(x)]^2} dx \\ &\approx \int_{x_a}^{x_b} \sqrt{1 + y_0'^2} \left[1 + \frac{1}{2} \left(\frac{2\epsilon \eta' y_0'}{1 + y_0'^2} \right) \right] dx. \end{aligned}$$

Only the linear term in ϵ are kept because ϵ is infinitesimally small. We shall call the first term to be L_0 and the second δL . If y_0 is an extremal path, the change in the path length from $L_0 \rightarrow L_0 + \delta L$ should have $\delta L = 0$ for all ϵ and η ,

$$\begin{aligned} 0 = \delta L &= \int_{x_a}^{x_b} \sqrt{1 + y_0'^2} \frac{\epsilon \eta' y_0'}{1 + y_0'^2} dx \\ &= \frac{\epsilon y_0'}{\sqrt{1 + y_0'^2}} \eta \Big|_{x_a}^{x_b} - \int_{x_a}^{x_b} \epsilon \eta \frac{d}{dx} \left[\frac{y_0'}{\sqrt{1 + y_0'^2}} \right] dx. \end{aligned}$$

The first term vanishes because there should be no deviation at the end points, $\eta(x_a) = \eta(x_b) = 0$. To make the second term zero for all ϵ and η , we have

$$\boxed{\frac{d}{dx} \left(\frac{y_0'}{\sqrt{1 + y_0'^2}} \right) = 0.}$$

If you solve this, you will find that $y'_0 = \text{const.}$, or y_0 is a straight line.

1.2 The Euler-Lagrange Equation

For a more general case, consider a function $F(y(x), y'(x), x)$ and we are to extremize

$$I = \int_{x_a}^{x_b} F(y(x), y'(x), x) dx$$

with the constraint $y(x_a) = y_a$ and $y(x_b) = y_b$, the endpoints fixed. Similar to the example in the previous section, let $y = y_0(x) + \epsilon\eta(x)$.

Then

$$\delta I = I[y_0 + \epsilon\eta] - I[y_0] = \int_{x_a}^{x_b} F(y_0 + \epsilon\eta, y'_0 + \epsilon\eta', x) dx - \int_{x_a}^{x_b} F(y_0, y'_0, x) dx.$$

Take ϵ to be infinitesimal and Taylor expand F ,

$$\delta I = \int_{x_a}^{x_b} \left[\frac{\partial F}{\partial(y_0 + \epsilon\eta)} \Big|_{\epsilon=0} \epsilon\eta + \frac{\partial F}{\partial(y'_0 + \epsilon\eta')} \Big|_{\epsilon=0} \epsilon\eta' \right] dx = \int_{x_a}^{x_b} \left(\epsilon\eta \frac{\partial F}{\partial y} + \epsilon\eta' \frac{\partial F}{\partial y'} \right) dx.$$

Requiring $\delta I = 0$ and integrating by parts to get rid of η' ,

$$\begin{aligned} 0 = \delta I &= \int_{x_a}^{x_b} \epsilon\eta \frac{\partial F}{\partial y} dx + \epsilon\eta \frac{\partial F}{\partial y'} \Big|_{x_a}^{x_b} - \int_{x_a}^{x_b} \epsilon\eta \frac{d}{dx} \frac{\partial F}{\partial y'} dx \\ &= \int_{x_a}^{x_b} \epsilon\eta \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right) dx. \end{aligned}$$

This works for arbitrary η , so

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0.} \quad (1.2)$$

This is known as the **Euler-Lagrange equation**. Calculus of variations reduces a problem of extremizing an integral to solving a differential equation, which is much more familiar with doing.

1.3 Two more examples

1.3.1 Fermat's Principle and Snell's law

Fermat's principle says that light traveling from two point A and B will choose the path that takes the shortest time. The path of light in a uniform medium is a straight line. However, if the medium is composed of two or more materials, the path is no longer trivial. We know that light travels with a speed of c/n in a material, where n represents the index of refraction. Light would “be in favor of” one medium than the other in this case.

Suppose we want to know the path of a light ray through the Earth's atmosphere, which is more dense closer to the Earth's surface. The index of refraction is a function of the height y , denoted by $n(y)$. We should find the path that minimize the traveling time of light using calculus of variations. Traveling a distance $ds = \sqrt{dx^2 + dy^2}$ with speed $c/n(y)$ takes time

$$dt = \frac{ds}{c/n(y)} = \frac{1}{c} n(y) \sqrt{1 + y'^2} dx = \frac{1}{c} n(y) \sqrt{1 + x'^2} dy.$$

We will use y as the independent variable for convenience. The total time taken is

$$t = \int_{y_a}^{y_b} \frac{1}{c} n(y) \sqrt{1 + x'^2} dy.$$

By the Euler-Lagrange equation (1.2),

$$0 = \frac{\partial F}{\partial x} - \frac{d}{dy} \frac{F}{x'} = 0 - \frac{d}{dy} \frac{n(y)x'}{c\sqrt{1+x'^2}} \implies \frac{n(y)x'}{c\sqrt{1+x'^2}} = C.$$

The full function of $x(y)$ cannot be determined unless $n(y)$ is specified, but we can get some other conclusions. By definition of the slope,

$$x'(y) = \frac{dx}{dy} = \tan \theta(y),$$

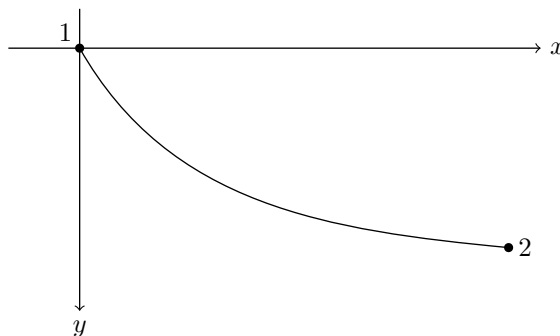
where $\theta(y)$ is the angle of the light ray relative to vertical (or normal). By some trigonometry, we have

$$\frac{x'}{\sqrt{1+x'^2}} = \sin \theta(y) \implies \boxed{n(y) \sin \theta(y) = C.}$$

We recover the [Snell's law](#), or the law of refraction.

1.3.2 Brachistochrone

The [brachistochrone](#) (“shortest time”) problem asks: given two points 1 and 2, with 1 higher above the ground, in what shape should we build a frictionless path so that an object released from point 1 will reach point 2 in the shortest possible time?



The time to travel from 1 to 2 is

$$t(1 \rightarrow 2) = \int_{y_1}^{y_2} \frac{ds}{v}$$

where the speed at any height y is determined by conservation of energy to be $v = \sqrt{2gy}$. Take y as the independent variable. Then the time of interest is

$$t = \frac{1}{\sqrt{2g}} \int_0^{y_2} \sqrt{\frac{x'^2 + 1}{y}} dy$$

with the integrand

$$f(x, x', y) = \frac{\sqrt{x'^2 + 1}}{\sqrt{y}}.$$

Applying the Euler-Lagrange equation (1.2),

$$\frac{x'^2}{y(1+x'^2)} = C = \frac{1}{2a}$$

where the constant $1/2a$ is for future convenience. This equation can be changed into

$$x' = \sqrt{\frac{y}{2a-y}} \implies x = \int \sqrt{\frac{y}{2a-y}} dy.$$

Substitute $y = a(1 - \cos \theta)$,

$$x = a \int \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \sin \theta d\theta = a \int \sqrt{\frac{(1 - \cos \theta)^2}{1 - \cos^2 \theta}} \sin \theta d\theta = a \int (1 - \cos \theta) d\theta = a(\theta - \sin \theta) + C_1$$

The initial point 1 have $x = y = 0$, so the initial value of θ is 0 and $C_1 = 0$. Now the parametric equation is

$$\begin{cases} x = a(\theta - \sin \theta), \\ y = a(1 - \cos \theta). \end{cases}$$

The path is a *cycloid*.

1.4 Relating to Mechanics

Paths may not be written in the form $y = y(x)$ in two dimension and for higher dimension spaces, we must use parametric forms,

$$x = x(t), \quad y = y(t), \quad z = z(t), \dots$$

where t is any convenient variable in terms of which the curve can be parameterized. The integral we might want to construct will take the form

$$S = \int_{t_1}^{t_2} F[x, y, z, \dots, \dot{x}, \dot{y}, \dot{z}, t] dt.$$

In this case, we will have multiple Euler-Lagrange equations:

$$\frac{\partial F}{\partial x} - \frac{d}{dt} \frac{\partial F}{\partial \dot{x}} = 0, \quad \frac{\partial F}{\partial y} - \frac{d}{dt} \frac{\partial F}{\partial \dot{y}} = 0, \quad \frac{\partial F}{\partial z} - \frac{d}{dt} \frac{\partial F}{\partial \dot{z}} = 0, \dots \implies \boxed{\frac{\partial F}{\partial q_i} - \frac{d}{dt} \frac{\partial F}{\partial \dot{q}_i} = 0.}$$

2 LAGRANGIAN MECHANICS

2.1 Lagrange Equations

2.1.1 Hamilton's Principle

Let us first consider a free particle with kinetic energy $T = \frac{1}{2}m\dot{\mathbf{r}}^2$. The action on this kinetic energy is

$$S = \int_{t_a}^{t_b} T dt = \int_{t_a}^{t_b} \frac{1}{2}m\dot{\mathbf{r}}^2 dt.$$

Extremizing the action gives three Euler-Lagrangian equations (1.2):

$$\frac{\partial T}{\partial x} - \frac{d}{dt} \frac{\partial T}{\partial \dot{x}} = 0, \quad \frac{\partial T}{\partial y} - \frac{d}{dt} \frac{\partial T}{\partial \dot{y}} = 0, \quad \frac{\partial T}{\partial z} - \frac{d}{dt} \frac{\partial T}{\partial \dot{z}} = 0.$$

Since the kinetic energy does not depend explicitly on position (x, y, z) but only on $(\dot{x}, \dot{y}, \dot{z})$, the first term in each equation vanishes. Combining the second terms of the three equations gives

$$\frac{d}{dt}(m\dot{\mathbf{r}}) = 0.$$

We see that extremizing the action S produces Newton's second law for a free particle. Then what if the particle is not free, but subject to some (conservative) force, or equivalently, some potential energy $V(x, y, z)$. Notice that the first term in the Euler-Lagrangian equation is a spatial derivatives. This may remind you of the relation between a conservative force and a potential energy,

$$\frac{\partial V}{\partial x} = -F_x, \quad \frac{\partial V}{\partial y} = -F_y, \quad \frac{\partial V}{\partial z} = -F_z.$$

What can we do to the action (or the integrand) to produce such a definition? Or particularly,

$$\mathbf{F} = -\nabla V = \frac{d}{dt}(m\dot{\mathbf{r}}).$$

The answer is: let the **action** be defined as

$$S = \int_{t_a}^{t_b} \mathcal{L} dt \tag{2.1}$$

where the integrand $\mathcal{L} = T - V$ is called the **Lagrangian**. Extremizing the action produces the Newton's second law, for a conservative force for now. Note that the independent variable in Lagrangian mechanics is the time t . The dependent variables are the coordinates that specify the position, or "configuration" of a system, and are usually denoted by $q_1, q_2, \dots, q_n$. For a single particle moving unconstrained in three dimensions, n is 3, and the three coordinates q_1, q_2, q_3 could be just the three Cartesian coordinates x, y, z , or they might be the spherical polar coordinates r, θ, ϕ .

The ultimate goal in most problems in Lagrangian mechanics is to find how the coordinates vary with time (to find the n functions $q_1(t), \dots, q_n(t)$), a path in the n -dimensional configuration space. This path can be determined by Newton's second law and equivalently, or can be characterized as the path for which a certain integral is stationary. This means that it must satisfy the

corresponding Euler-Lagrange equations (called just Lagrange equations in this context). There are two core differences in these two formulations

- Newtonian mechanics often formulate the problem as a “time-evolution problem”. With initial condition of a particle specified, one can know the position, velocity, acceleration and so on, by solving Newton’s second law. Lagrangian mechanics, on the other hand, formulate the problem as a “path problem”. With the initial position and final position specified, one can find the path of a particle in which it must follow physically. Both are formulations of physical real-world problems.
- Lagrangian formulation has an advantage over Newtonian one. In particular, unlike Newton’s second law, Lagrange’s equations take exactly the same simple form in *all* coordinate systems. This is because the action is a scalar quantity and it is frame invariant. What’s going on *physically* is always the same in all frame of reference. It will be painful to write down Newton’s second law in non-Cartesian coordinates (e.g. spherical coordinates), because there are non-trivial derivatives on the unit vectors. For example, consider a free particle in two dimensional polar coordinates. To get its equation of motion in polar coordinates, we use the Lagrangian

$$\mathcal{L} = T - V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) - 0.$$

The Lagrange equations give

$$0 = \frac{\partial T}{\partial r} - \frac{d}{dt} \frac{\partial T}{\partial \dot{r}} = m\dot{\phi}^2 - m\ddot{r}, \quad 0 = -\frac{\partial T}{\partial \phi} - \frac{d}{dt} \frac{\partial T}{\partial \dot{\phi}} = -\frac{d}{dt}(mr^2\dot{\phi}) = -2mr\dot{\phi} - mr^2\ddot{\phi}.$$

Another advantage of Lagrangian formulation is that it handles **constraints** on particles effectively. For a particle with constraints, one can choose a **generalized coordinates** q_i that are sufficient to specify where every object is. We will have several examples and a proof on this topic in following sections.

- One disadvantage of Lagrangian mechanics is that it is hard to incorporate friction or other dissipating forces. These forces are easier to include in Newtonian mechanics.

In short, the take away from this section is what is called **Hamilton’s principle** (or sometimes **principle of least action** because for most of the time, action is minimized): the actual path which a particle follows between two points 1 and 2 in a given time interval, t_1 and t_2 , is such that the action is stationary,

$$\delta S = \delta \int_{t_1}^{t_2} \mathcal{L} dt = 0 \quad (2.2)$$

2.1.2 Non-Uniqueness of the Lagrangian

Theorem 2.1. Consider a Lagrangian $\mathcal{L}' = \mathcal{L} + df/dt$, where the Lagrangian $\mathcal{L} = \mathcal{L}(q_i, \dot{q}_i, t)$ obeys Lagrange equations and the function $f = f(q_i, t)$ does not depends on $\dot{q}_i$. Then $\mathcal{L}'$ produces the same equations of motion as $\mathcal{L}$. In other words, the Lagrangian of a system is not unique.

We shall first show that the new Lagrangian $\mathcal{L}'$ depends on the correct variables. Take the time derivative of f ,

$$\frac{df}{dt} = \frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial t}.$$

Substitute it into $\mathcal{L}'$,

$$\mathcal{L}' = \mathcal{L}(q_i, \dot{q}_i, t) + \frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial t} = \mathcal{L}'(q_i, \dot{q}_i, t).$$

Note that if f depends on $\dot{q}_i$, then its time derivative would contain $\ddot{q}_i$ in general, and so would $\mathcal{L}'$. This is why we need $f = f(q_i, t)$. Now if $\mathcal{L}$ obeys Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i}.$$

For $\mathcal{L}'$,

$$\begin{aligned} \frac{\partial \mathcal{L}'}{\partial q_i} - \frac{d}{dt} \frac{\partial \mathcal{L}'}{\partial \dot{q}_i} &= \frac{\partial}{\partial q_i} \left(\mathcal{L} + \frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial t} \right) - \frac{d}{dt} \frac{\partial}{\partial \dot{q}_i} \left(\mathcal{L} + \frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial t} \right) \\ &= \frac{\partial \mathcal{L}}{\partial q_i} + \frac{\partial^2 f}{\partial q_i^2} \dot{q}_i + \frac{\partial^2 f}{\partial q_i \partial t} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} + \frac{\partial f}{\partial q_i} \right) \\ &= \left(\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) + \frac{\partial^2 f}{\partial q_i^2} \dot{q}_i + \frac{\partial^2 f}{\partial q_i \partial t} - \frac{d}{dt} \frac{\partial f}{\partial q_i} \end{aligned}$$

The first term is just zero according to Lagrange equations, and we can expand the last total time derivative by product rule,

$$\frac{\partial \mathcal{L}'}{\partial q_i} - \frac{d}{dt} \frac{\partial \mathcal{L}'}{\partial \dot{q}_i} = 0 + \frac{\partial^2 f}{\partial q_i^2} \dot{q}_i + \frac{\partial^2 f}{\partial q_i \partial t} - \frac{\partial^2 f}{\partial q_i^2} \dot{q}_i - \frac{\partial}{\partial t} \frac{\partial f}{\partial q_i} = 0.$$

Thus, $\mathcal{L}'$ also obeys Lagrange equations. $\mathcal{L}$ and $\mathcal{L}'$ should produce the same (or at least equivalent) equations of motion.

2.2 Proof and Examples of Lagrange's Equations with Constraints

2.2.1 Proof

Consider a particle that is constrained to move on a surface. This means that it has two degrees of freedom and can be described by two generalized coordinates q_1 and q_2 that can vary independently.

There are two kinds of forces on the particle. First, there are the forces of constraints. In general, the forces of constraint are not necessarily conservative, but this does not matter. The constraining force on the particle is denoted by $\mathbf{F}_{\text{cstr}}$, which in this case is the normal force of the surface to which the particle is confined. In real world, all the constraint forces are normal to the velocity of a particle, which means they do no work.

Second, there are all other unconstrained but conservative forces on the particle, such as gravity. These forces are denoted by $\mathbf{F}$. Assume that the nonconstraint forces are derivable from a potential energy, $V(\mathbf{r}, t)$, and $\mathbf{F} = -\nabla V$. The total force on the particle is $\mathbf{F}_{\text{tot}} = \mathbf{F}_{\text{cstr}} + \mathbf{F}$.

Consider the initial and final points $\mathbf{r}_1$ and $\mathbf{r}_2$ through which the particle passes between time t_1 and t_2 . Denote $\mathbf{r}(t)$ the “right” path, the actual path that the particle follows between the two points, and by $\mathbf{R}(t)$ any neighboring “wrong” path between the same two points.

$$\mathbf{R}(t) = \mathbf{r}(t) + \boldsymbol{\epsilon}(t),$$

which defined $\boldsymbol{\epsilon}(t)$ as the infinitesimal vector pointing from $\mathbf{r}(t)$ on the right path to the corresponding point $\mathbf{R}(t)$ on the wrong path. Since both $\mathbf{r}(t)$ and $\mathbf{R}(t)$ go through the same endpoints, $\boldsymbol{\epsilon}(t_1) = \boldsymbol{\epsilon}(t_2) = 0$. The action

$$S = \int_{t_1}^{t_2} \mathcal{L}(\mathbf{R}, \dot{\mathbf{R}}, t) dt$$

is taken along any path $\mathbf{R}(t)$ lying in the constraining surface, and S_0 is the corresponding action taken along the right path $\mathbf{r}(t)$. Now is to prove that the action S is stationary for variations of the path $\mathbf{R}(t)$, when $\mathbf{R}(t) = \mathbf{r}(t)$ or, equivalently, when the difference

$\epsilon = 0$. Another way to say this is that the difference in the action

$$\delta S = S - S_0$$

is zero to first order in the distance ϵ between the paths. The difference between the Lagrangians on the two paths is

$$\delta \mathcal{L} = \mathcal{L}(\mathbf{R}, \dot{\mathbf{R}}, t) - \mathcal{L}(\mathbf{r}, \dot{\mathbf{r}}, t).$$

Substitute $\mathbf{R}(t) = \mathbf{r}(t) + \epsilon(t)$ and

$$\mathcal{L}(\mathbf{r}, \dot{\mathbf{r}}, t) = T - V = \frac{1}{2} m \dot{\mathbf{r}}^2 - V(\mathbf{r}, t).$$

The difference of the Lagrangians becomes

$$\delta \mathcal{L} = \frac{1}{2} m [(\dot{\mathbf{r}} + \dot{\epsilon})^2 - \dot{\mathbf{r}}^2] - [V(\mathbf{r} + \epsilon, t) - V(\mathbf{r}, t)] = m \dot{\mathbf{r}} \cdot \dot{\epsilon} - \epsilon \cdot \nabla V + \mathcal{O}(\epsilon^2).$$

Then the difference in action is

$$\begin{aligned} \delta S &= \int_{t_1}^{t_2} \delta \mathcal{L} dt = \int_{t_1}^{t_2} [m \dot{\mathbf{r}} \cdot \dot{\epsilon} - \epsilon \cdot \nabla V] dt \\ &= m \dot{\mathbf{r}} \cdot \epsilon \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \epsilon \cdot [m \ddot{\mathbf{r}} + \nabla V] dt \\ &= - \int_{t_1}^{t_2} \epsilon \cdot [m \ddot{\mathbf{r}} + \nabla V] dt. \end{aligned}$$

The term $m \ddot{\mathbf{r}}$ is the total force $\mathbf{F}_{\text{tot}}$ on the particle, while $\nabla V = -\mathbf{F}$. Therefore,

$$\delta S = - \int_{t_1}^{t_2} \epsilon \cdot \mathbf{F}_{\text{cstr}} dt.$$

The constraint force $\mathbf{F}_{\text{cstr}}$ is normal to the surface in which the particle moves, while ϵ lies in the surface, so $\epsilon \cdot \mathbf{F}_{\text{cstr}} = 0$.

The particle is confined to move on a two dimensional subset of the full three-dimensional world. This means that the particle has two degrees of freedom and can be described by two generalized coordinates, q_1 and q_2 . Any variation of q_1 and q_2 is consistent with the constraints.

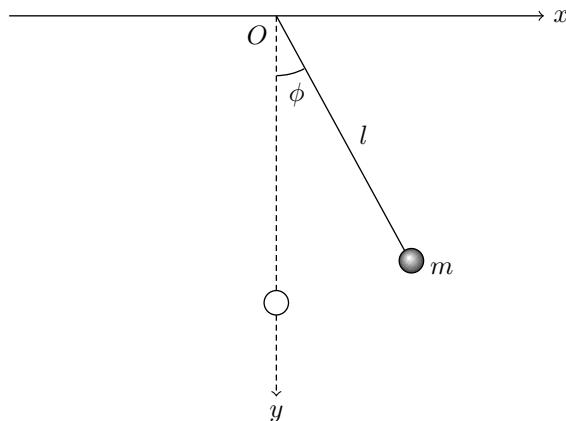
The principle of least action and action itself imply that

$$\frac{\partial \mathcal{L}}{\partial q_1} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_1} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial q_2} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_2}.$$

The proof here applies directly only to a single particle in three dimensions, constrained to move on a two-dimensional surface, but the main ideas of the general case are all present,

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i}.$$

2.2.2 Example: Pendulum



Consider the simple pendulum shown above. A bob of mass m is fixed to a massless rod, which is pivoted at O and free to swing without friction in the xy plane. The bob moves in both the x and y directions, but it is constrained by the rod so that $\sqrt{x^2 + y^2} = l$ remains constant. A simpler way to express both x and y is in terms of a single parameter ϕ , the angle between the pendulum and its equilibrium position. The kinetic energy and potential energy are

$$T = \frac{1}{2}mv^2 = \frac{1}{2}ml^2\dot{\phi}^2 \quad \text{and} \quad V = mgh$$

where $h = l(1 - \cos \phi)$. The Lagrangian is

$$\mathcal{L} = T - V = \frac{1}{2}ml^2\dot{\phi}^2 - mgl(1 - \cos \phi).$$

Now the Lagrange's equation reads

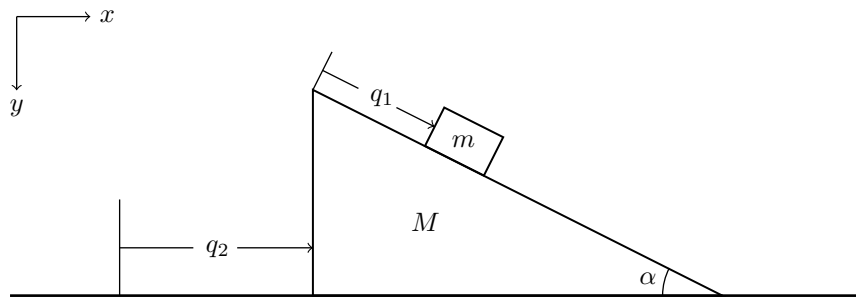
$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \quad \Rightarrow \quad -mgl \sin \phi = \frac{d}{dt} (ml^2 \dot{\phi}) = ml^2 \ddot{\phi}.$$

The LHS of this equation is just the torque τ exerted by gravity on the pendulum, while the term ml^2 is the pendulum's moment of inertia I . Since $\ddot{\phi}$ is the angular acceleration α , we see that Lagrange's equation for the simple pendulum simply reproduces the familiar result $\tau = I\alpha$.

2.2.3 Example: A Block Sliding on a Wedge

Consider a block and wedge shown below. The block (mass m) is free to slide on the wedge, and the wedge (mass M) can *also* slide on the horizontal table, both with negligible friction. The block is released from the top of the wedge, with both initially at rest. If the wedge has angle α and the length of its sloping face is l , how long does the block take to reach the bottom?

A good choice of the two generalized coordinates is, the distance q_1 of the block from the top of the wedge and the distance q_2 of the wedge from any convenient fixed point on the table. The quantity we need to find is the acceleration $\ddot{q}_1$ of the block relative to the wedge.



The kinetic energy of the wedge is simply $T_M = \frac{1}{2}M\dot{q}_2^2$. The velocity of the block relative to the table is

$$\mathbf{v} = (v_x, v_y) = (\dot{q}_1 \cos \alpha + \dot{q}_2, \dot{q}_1 \sin \alpha).$$

Thus, the kinetic energy of the block is

$$T_m = \frac{1}{2}m(v_x^2 + v_y^2) = \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2 \cos \alpha).$$

The total kinetic energy of the system is

$$T = T_M + T_m = \frac{1}{2}(M + m)\dot{q}_2^2 + \frac{1}{2}m(\dot{q}_1^2 + 2\dot{q}_1\dot{q}_2 \cos \alpha).$$

The potential energy of the wedge is a constant, so take it to be 0. That of the block is $-mgy$, where $y = q_1 \sin \alpha$. Therefore,

$$V = -mgq_1 \sin \alpha.$$

The Lagrangian is

$$\mathcal{L} = T - V = \frac{1}{2}(M + m)\dot{q}_2^2 + \frac{1}{2}m(\dot{q}_1^2 + 2\dot{q}_1\dot{q}_2 \cos \alpha) + mgq_1 \sin \alpha.$$

The q_2 equation is

$$\frac{\partial \mathcal{L}}{\partial q_2} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_2} \implies M\dot{q}_2 + m(\dot{q}_2 + \dot{q}_1 \cos \alpha) = \text{const.}$$

This is the conservation of the total momentum in the x direction.

The q_1 equation is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial q_1} &= \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_1} \implies mg \sin \alpha = \frac{d}{dt} m(\dot{q}_1 + \dot{q}_2 \cos \alpha) \\ &= m(\ddot{q}_1 + \ddot{q}_2 \cos \alpha). \end{aligned}$$

Differentiating the q_2 equation,

$$\ddot{q}_2 = -\frac{m}{M + m}\ddot{q}_1 \cos \alpha.$$

Eliminate $\ddot{q}_2$ and solve for $\ddot{q}_1$:

$$\ddot{q}_1 = \frac{g \sin \alpha}{1 - \frac{m \cos^2 \alpha}{M + m}}.$$

The acceleration down the slope is constant. The time to travel the length l is just $\sqrt{2l/\ddot{q}}$. We may check that $\ddot{q}_1$ agrees with common sense in various special cases:

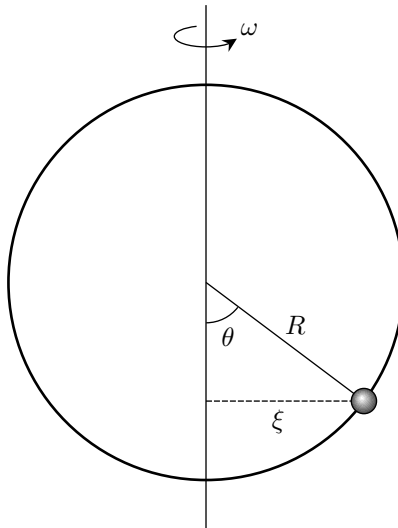
- If $\alpha = 90^\circ$, $\ddot{q}_1 = g$.
- If $M \rightarrow \infty$, $\ddot{q}_1 \rightarrow g \sin \alpha$, which is the case of treating the wedge as fixed incline.

- If $M \rightarrow 0$, $\ddot{q}_1 = g/\sin \alpha$, which means the y component of this acceleration is $\ddot{q}_1 \sin \alpha = g$.

2.2.4 A Bead on a Spinning Hoop

A bead of mass m is threaded on a frictionless circular wire hoop of radius R . The hoop lies in a vertical plane, which is forced to rotate about the hoop's vertical diameter with constant angular velocity $\dot{\phi} = \omega$, as shown in the figure. The bead's position on the hoop is specified by the angle θ measured up from the vertical.

Write down the Lagrangian for the system in terms of the generalized coordinate θ and find the equation of motion of the bead. Find any equilibrium position at which the bead can remain with θ constant. Use the equation of motion to discuss the stability of the equilibrium positions.



Relative to a nonrotating frame, the bead has velocity $R \cdot \dot{\theta}$ tangential to the hoop and $\xi\omega = (R \sin \theta)\omega$ normal to the hoop. The kinetic energy is

$$T = \frac{1}{2}mv^2 = \frac{1}{2}mR^2 (\dot{\theta}^2 + \omega^2 \sin^2 \theta).$$

The gravitational potential energy is

$$U = mgR(1 - \cos \theta).$$

The Lagrangian is

$$\mathcal{L} = \frac{1}{2}mR^2 (\dot{\theta}^2 + \omega^2 \sin^2 \theta) - mgR(1 - \cos \theta),$$

and the Lagrange equation is

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} \implies mR^2 \omega^2 \sin \theta \cos \theta - mgR \sin \theta = mR^2 \ddot{\theta}.$$

Dividing through by mR^2 ,

$$\ddot{\theta} = (\omega^2 \cos \theta - g/R) \sin \theta. \quad (2.3)$$

An equilibrium point is any value of θ , called θ_0 , satisfying the following condition: if the bead is placed at rest ($\dot{\theta} = 0$) at $\theta = \theta_0$, then it will remain at rest at θ_0 . The condition is guaranteed if $\ddot{\theta} = 0$.

To find equilibrium positions, we must have the RHS of (2.3) equal to 0:

$$(\omega^2 \cos \theta - g/R) \sin \theta = 0.$$

Thus, $\theta = 0$ or π , the bead remain at rest at the bottom or top of the hoop; another case is

$$\cos \theta = \frac{g}{\omega^2 R}.$$

which can vanish only when $\omega^2 \geq g/R$, so there are two more equilibrium positions at

$$\theta_0 = \pm \arccos\left(\frac{g}{\omega^2 R}\right).$$

An equilibrium point θ_0 is not especially interesting unless it is stable. That is, the bead will move back toward θ_0 if it is nudged a little away from θ_0 . Now let's analyze some cases:

1. At the bottom, $\theta = 0$, as long as θ remains close to 0, we can set $\cos \theta \approx 1$ and $\sin \theta \approx \theta$ and approximate the equation by

$$\ddot{\theta} = (\omega^2 - g/R)\theta \quad [\theta \text{ near } 0].$$

If the hoop is rotating slowly ($\omega^2 < g/R$),

$$\ddot{\theta} = (\text{negative number})\theta.$$

If we nudge the bead, the bead returns toward the equilibrium, which is *stable*. If speed up the rotation of the hoop, so that $\omega^2 > g/R$, then

$$\ddot{\theta} = (\text{positive number})\theta.$$

The equilibrium at the bottom changes from stable to unstable.

2. The equilibrium at the top ($\theta = \pi$) is always unstable.
3. The other two equilibrium positions only exist when $\omega^2 > g/R$. The equation of motion 2.3 is

$$\ddot{\theta} = (\omega^2 \cos \theta - g/R) \sin \theta.$$

At the equilibrium point, the term in parenthesis is 0, while $\sin \theta$ is positive. If we increase θ a little, $\sin \theta$ remains positive, but the term in parenthesis becomes negative. Thus $\ddot{\theta}$ becomes negative, and the bead accelerates back toward its equilibrium position. In other direction, the result is the same, the equilibrium position is *stable*.

Interestingly, when the hoop is rotating slowly ($\omega^2 < g/R$), there is just one stable equilibrium, at $\theta = 0$. If we speed up the rotation, then as ω passes the critical value where $\omega^2 = g/R$, this original equilibrium becomes unstable, but two new stable equilibrium points appear, emerging from $\theta = 0$ and moving out to the right and left as we increase ω still more. This phenomenon—the disappearance of one stable equilibrium and the simultaneous appearance of two others diverging from the same point—is called a **bifurcation**.

2.3 Conserved Quantities and Symmetries

2.3.1 Generalized Momentum

A conservation law is the statement that some quantity Q is independent of time with sufficient initial conditions

$$\frac{dQ}{dt} = 0.$$

In the Lagrange equation,

$$0 = \frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i},$$

if $\partial \mathcal{L} / \partial q_i = 0$ for all time, then $\partial \mathcal{L} / \partial \dot{q}_i$ is a conserved quantity. This is saying that q_i is completely absent from the Lagrangian, i.e., it is a **cyclic** or **ignorable coordinate**. If we use Cartesian coordinates and $T = \frac{1}{2} m \dot{x}^2$, then $\partial \mathcal{L} / \partial \dot{x} = p_x$, the momentum, is conserved. We can define the **generalized momentum** as

$$p_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{q}_i}.$$

If a generalized coordinate is cyclic, then the corresponding generalized momentum is conserved. With this definition, we can say that $\partial \mathcal{L} / \partial q_i$ is the **generalized force**: it is the time-derivative of the generalized momentum.

Example 2.1. Angular Momentum

Consider a spherical pendulum: a mass attached to a massless, unstretchable string of length R pivoted at some point. The pendulum and the string can move under the influence of gravity. It is a three dimensional problem, but since r is fixed to be R , then we only need two generalized coordinates. It is convenient to choose the ordinary spherical coordinates ϕ, θ as our generalized coordinates. The Lagrangian is

$$\mathcal{L} = T - V = \frac{1}{2} m R^2 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) - mgR(1 - \cos \theta).$$

Note that there is no explicit dependence of ϕ in the Lagrangian, so it is a cyclic coordinate. The corresponding generalized momentum is

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = m R^2 \dot{\phi} \sin^2 \theta.$$

You may recognize p_ϕ is the angular momentum around the z -axis. If we proceed with Lagrange equations:

$$\begin{aligned} 0 &= \frac{\partial \mathcal{L}}{\partial \theta} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m R^2 \dot{\phi}^2 \sin \theta \cos \theta - mgR \sin \theta - m R^2 \ddot{\theta}, \\ 0 &= \frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = -2m R^2 \dot{\theta} \dot{\phi} \sin \theta \cos \theta - m R^2 \ddot{\phi} \sin^2 \theta, \end{aligned}$$

we need to solve a system of second-order nonlinear equations, which are in general extremely hard to solve.

However, if we use the fact that p_ϕ is conserved, we can substitute p_ϕ into the first equation

$$0 = \frac{\cos \theta}{m R^2 \sin^3 \theta} p_\phi^2 - mgR \sin \theta - m R^2 \ddot{\theta}.$$

Then treat p_ϕ as a constant determined by initial conditions, we have one equation only in θ .

It can be shown that the Lagrange equation can be rewritten in the form

$$0 = \frac{\partial \mathcal{L}}{\partial t} + \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{q}_i - \mathcal{L} \right), \quad (2.4)$$

where Einstein summation notation is used for the second term. If the Lagrangian does not explicitly depend on time, the quantity $p_i \dot{q}_i - \mathcal{L}$ is conserved. This quantity is called the **Hamiltonian**,

$$H = p_i \dot{q}_i - \mathcal{L}. \quad (2.5)$$

In Cartesian coordinates and assume that the potential energy is a function of position, the Hamiltonian is the total energy of the system,

$$H = T + V.$$

It turns out that as long as the transformation between Cartesian coordinates and the generalized coordinates does not depend explicitly on time, the Hamiltonian is just the energy.

2.3.2 Noether's Theorem

The conserved quantities are closely connected to symmetries in the Lagrangian. Consider an infinitesimal transformation in the generalized coordinates

$$q'_i = q_i + \delta q_i(q_i, \dot{q}_i, t) = q_i + \epsilon \xi_i(q_i, \dot{q}_i, t)$$

for $\epsilon \ll 1$.

For example, a translation in the z -direction will have a transformation

$$\delta x = 0, \quad \delta y = 0, \quad \delta z = \epsilon z_0,$$

where z_0 is some finite constant with units of length. Another example could be the rotation around the z -axis. For a rotation of ϕ around the azimuth,

$$x' = x \cos \phi - y \sin \phi, \quad y' = x \sin \phi + y \cos \phi, \quad z' = z.$$

If we want the infinitesimal rotation, take $\phi \ll 1$, and

$$x' = x - \phi y, \quad y' = \phi x + y, \quad z' = z.$$

In this case, the transformation in the coordinates is $(\xi_x = -y, \xi_y = x, \xi_z = 0)$. This is an example where the change in coordinates depends on position.

For any transformation, the change in action can be written as

$$\begin{aligned} \delta S &= S[q'] - S[q] \\ &= \int_{t_a}^{t_b} [\mathcal{L}(q'_i, \dot{q}'_i, t) - \mathcal{L}(q_i, \dot{q}_i, t)] \\ &= \int_{t_a}^{t_b} [\mathcal{L}(q_i + \epsilon \xi_i, \dot{q}_i + \epsilon \dot{\xi}_i, t) - \mathcal{L}(q_i, \dot{q}_i, t)] \\ &= \int_{t_a}^{t_b} \left(\frac{\partial \mathcal{L}}{\partial q_i} \epsilon \xi_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \epsilon \dot{\xi}_i \right) dt. \end{aligned}$$

The integrand is just $\delta \mathcal{L}$, expressed in multivariable Taylor expansion around $\epsilon = 0$. This integral is in general not zero, but with an appropriate choice of ξ_i , we can have

$$\delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial q_i} \epsilon \xi_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \epsilon \dot{\xi}_i = 0.$$

Such a transformation is called a **symmetry**. By this definition, $\delta \mathcal{L}$ needs to vanish at every value of t and no matter what the trajectory $q_i(t)$ is.

Suppose $q_i(t)$ is an actual physical trajectory that solves the Lagrange's equation,

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i}.$$

Substitute this into $\delta \mathcal{L}$, we have

$$\delta \mathcal{L} = \epsilon \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \xi_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{\xi}_i \right] = \epsilon \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \xi_i \right).$$

This is saying that whenever the given transformation ξ_k is a symmetry in the Lagrangian (or the action), the quantity $\xi_i \partial \mathcal{L} / \partial \dot{q}_i = p_i \xi_i$ is conserved along a physical trajectory. This conclusion is called **Noether's theorem**: for every symmetry, there is a corresponding conserved quantity. Here are some examples:

Symmetry	Conserved quantity
Translation	Linear momentum
Rotation	Angular momentum
Time	Hamiltonian

The translational symmetry is simple. If $\xi_z = z_0$, then we will have a conserved linear momentum in z -direction, $p_z \xi_z = p_z z_0$ (z_0 is just a constant). For rotational symmetry, the easiest way is of course to use spherical coordinates. For a transformation $\xi_\phi = \phi_0$, $p_\phi \phi_0$ is conserved, and recall that p_ϕ is just the angular momentum in the z -direction. If we stick to Cartesian coordinates and use the result derived previously ($\xi_x = -y, \xi_y = x, \xi_z = 0$), suppose this is a symmetry in the Lagrangian,

$$0 = \frac{\partial \mathcal{L}}{\partial q_i} \xi_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{\xi}_i = -y \frac{\partial \mathcal{L}}{\partial x} + x \frac{\partial \mathcal{L}}{\partial y} - \dot{y} \frac{\partial \mathcal{L}}{\partial \dot{x}} + \dot{x} \frac{\partial \mathcal{L}}{\partial \dot{y}}.$$

Then the conserved quantity is

$$p_i \xi_i = -y p_x + x p_y = L_z,$$

the angular momentum (expressed in Cartesian coordinates) in the z -direction as expected.

The case for the Hamiltonian requires some tricks. We already know that it is conserved when Lagrangian does not explicitly depend on time. Let the change in the Lagrangian be a total derivative of a function with respect to time,

$$\delta \mathcal{L} = \mathcal{L}(q'_i, \dot{q}'_i, t) - \mathcal{L}(q_i, \dot{q}_i, t) = \epsilon \frac{dK}{dt}.$$

We also know that for $q_i(t)$ satisfying the Lagrange equation has

$$\delta \mathcal{L} = \epsilon \left(\frac{\partial \mathcal{L}}{\partial q_i} \xi_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{\xi}_i \right) = \epsilon \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \xi_i \right).$$

This means

$$\frac{dK}{dt} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \xi_i \right) \implies \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \xi_i - K \right) = 0.$$

The function K depends on the choice of transformation. But we see that even though $\delta \mathcal{L}$ is not unchanged under this transformation, there can still be a conserved quantity $p_i \xi_i - K$. This conserved quantity looks a lot like the Hamiltonian, $H = p_i \dot{q}_i - \mathcal{L}$. Consider a transformation

$$\xi_i = \dot{q}_i t_0,$$

where t_0 is some constant with units of time. This transformation is like how the system evolve in time: $\delta q \sim \epsilon \xi_i = \dot{q}_i (\epsilon t_0)$. The

change in the Lagrangian is

$$\delta\mathcal{L} = \epsilon t_0 \frac{\partial\mathcal{L}}{\partial q_i} \dot{q}_i + \epsilon t_0 \frac{\partial\mathcal{L}}{\partial \dot{q}_i} \ddot{q}_i = \epsilon t_0 \left(\frac{\partial\mathcal{L}}{\partial q_i} \dot{q}_i + \frac{\partial\mathcal{L}}{\partial \dot{q}_i} \ddot{q}_i + \frac{\partial\mathcal{L}}{\partial t} - \frac{\partial\mathcal{L}}{\partial t} \right) = \epsilon t_0 \left(\frac{d\mathcal{L}}{dt} - \frac{\partial\mathcal{L}}{\partial t} \right).$$

If $\partial\mathcal{L}/dt = 0$, then

$$\delta\mathcal{L} = \frac{d(\epsilon t_0 \mathcal{L})}{dt} = \epsilon \frac{dK}{dt}, \quad \text{where } K \equiv t_0 \mathcal{L}.$$

We have the conserved quantity

$$p_i \xi_i - K = p_i \dot{q}_i t_0 - t_0 \mathcal{L} = t_0 (p_i \dot{q}_i \mathcal{L}).$$

This is the Hamiltonian up to a constant t_0 . In conclusion, the symmetry $\partial\mathcal{L}/dt = 0$ corresponds to conservation of the Hamiltonian.

3 TWO-BODY CENTRAL FORCE PROBLEMS

A central force between two objects is a force that only points towards or away from one object to the other. Such a force can be gravity, electrostatic force, spring force, etc. A two-body central force problem is a typical application of generalized coordinates and conserved quantities in Lagrangian mechanics.

3.1 The Generalized Coordinates and Lagrangian

Suppose there are two objects, specified by position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$. In total, there are six degrees of freedom. The Lagrangian in Cartesian coordinates is

$$\mathcal{L} = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 - V(\mathbf{r}_1, \mathbf{r}_2).$$

The potential energy is not an arbitrary function of the two positions. If the force is central, then the potential should depend only on the distance between two objects, $|\mathbf{r}_1 - \mathbf{r}_2|$. It is convenient to choose three of the generalized coordinates to be the separation vector between them,

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2.$$

Now the potential is $V(r)$. Since there are no external force on the two objects, we know the total momentum should be conserved. This suggests that we should choose (the other three) generalized coordinates such that the total momentum is the corresponding generalized momentum. This generalized coordinate is the position of the center of mass,

$$\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{M}, \quad \text{where } M = m_1 + m_2.$$

To see this, we can replace $\mathbf{r}_1$ and $\mathbf{r}_2$ with separation $\mathbf{r}$ and center of mass $\mathbf{R}$,

$$\mathbf{r}_1 = \mathbf{R} + \frac{m_2}{M}\mathbf{r}, \quad \mathbf{r}_2 = \mathbf{R} - \frac{m_1}{M}\mathbf{r}.$$

The Lagrangian in terms of $\mathbf{R}$ and $\mathbf{r}$ is

$$\mathcal{L} = \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}\mu\dot{\mathbf{r}}^2 - V(r),$$

where $\mu = m_1m_2/M$ is called the **reduced mass**. Now the Lagrangian is independent of $\mathbf{R}$ explicitly, so the corresponding generalized momentum $\partial\mathcal{L}/\partial\dot{\mathbf{R}} = M\dot{\mathbf{R}} = \mathbf{p}_{\text{total}}$ is conserved. Such choices of generalized coordinate reduce a six-dimensional problem to two three-dimensional problems, one for the center of mass motion, and the other for relative motion. Particularly, the center of mass motion is simple: it is moving in a straight line with constant velocity in absence of external forces. All we need to do is to solve the relative motion part.

We can analyze the relative motion part further before dive into calculations. Since the potential energy is a function of r , no matter how we rotate the system, the potential energy is the same. This is saying that the potential obeys rotational symmetry, so the total angular momentum (vector) is conserved. It tells us that the motion of the two objects is in a *plane*. We can choose spherical coordinates and let the motion lie in the plane of $\theta = \pi/2$. Note that it must be $\theta = \pi/2$ because other polar angle with fixed r

describes a cone.

Now we are left with a two dimensional problem, with the Lagrangian

$$\mathcal{L} = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\phi}^2) - V(r).$$

The Lagrangian is independent of ϕ , so we have a conserved quantity

$$L = p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \mu r^2 \dot{\phi}. \quad (3.1)$$

This is the total angular momentum in the CM (center of mass) frame, not the angular momentum of either particle. Moreover, the Lagrangian is independent of time, so the energy (or Hamiltonian) is also conserved,

$$E = H = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\phi}^2) + V(r). \quad (3.2)$$

To proceed further, there are many equations to use: conserved L equation, conserved E equation, and we haven't used any Lagrangian equations.

3.2 The First Approach

The first approach is to use conserved quantities. Substitute equation (3.1) into (3.2) gives

$$E = \frac{1}{2}\mu\dot{r}^2 + V_{\text{eff}}(r), \quad \text{where} \quad V_{\text{eff}} \equiv \frac{L^2}{2\mu r^2} + V(r). \quad (3.3)$$

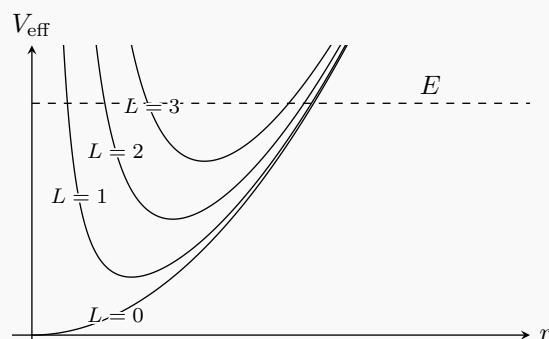
This is the final equation, a one-dimensional problem with a differential equation only of r . The conserved quantity L and E can be fixed by initial conditions. This equation has some physical implications: it looks like a particle of mass μ moving in a potential $V_{\text{eff}}(r)$ called the **effective potential**. Once we have solved $r(t)$, it is easy to compute $\phi(t)$,

$$\dot{\phi} = \frac{L}{\mu r(t)^2} \implies \phi(t) = \int_{t_0}^t dt' \frac{L}{\mu r(t')^2}.$$

Example 3.1. Harmonic oscillator potential

Consider the harmonic oscillator potential $V(r) = kr^2/2$, and so

$$V_{\text{eff}}(r) = \frac{L^2}{2\mu r^2} + \frac{kr^2}{2}.$$

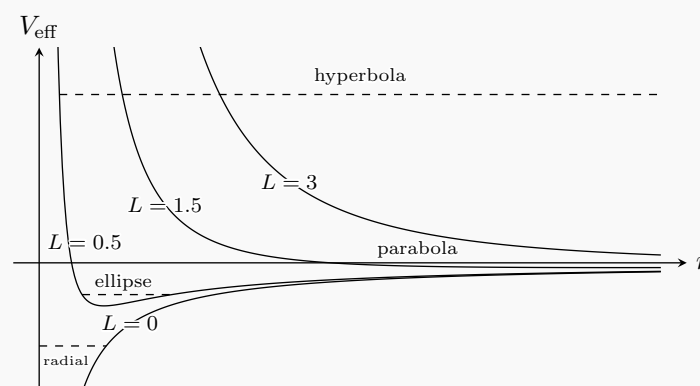


Above is a plot of V_{eff} with different angular momentum. A dashed line of fixed energy can be drawn. An object can only oscillate between the points where $E \geq V_{\text{eff}}$ because the kinetic energy $\mu\dot{r}^2/2$ cannot be negative. For a harmonic oscillator potential, there exist only bound states. For systems with nonzero L , they cannot get arbitrarily small r because of the $1/r^2$ term. This is called a **centrifugal barrier**.

Example 3.2. Gravitational potential

A gravitational potential is a $1/r$ potential,

$$V_{\text{eff}}(r) = \frac{L^2}{2\mu r^2} - \frac{Gm_1m_2}{r}.$$



The plot of V_{eff} is much more interesting in the $1/r$ case. With different values of energy, there exist two types of states. The type of orbits also depends on the energy.

If $E < 0$, the orbit is bound either radial ($L = 0$) or elliptical ($L \neq 0$).

- If $E = 0$, the orbit is unbound and parabolic.
- If $E > 0$, the orbit is unbound and hyperbolic.

The $E = 0$ line is just a convention. We can always shift the potential by some constant (and E should be shifted by the same constant). Qualitatively, a bound object cannot reach $r \rightarrow \infty$, while an unbound one can. Later we will discuss how mathematically obtain the equation of orbits. and they will appear to be conic sections (ellipse, parabola, and hyperbola).

3.3 The Second Approach

3.3.1 The Orbits

One may be interested the shape of the trajectories $r(\phi)$ instead of $r(t)$ and $\phi(t)$. What we do is to cancel the time derivative in the equation

$$E = \frac{1}{2}\mu \left(\dot{r}^2 + \frac{L^2}{\mu r^2} \right) + V(r) \quad \text{and} \quad L = \mu r^2 \dot{\phi}.$$

In order to do this, we solve for the time-derivative of r and ϕ ,

$$\dot{r} = \pm \sqrt{\frac{2E}{\mu} - \frac{2V}{\mu} - \frac{L^2}{\mu^2 r^2}}, \quad \dot{\phi} = \frac{L}{\mu r^2}.$$

Then the derivative of r with respect to ϕ is

$$r' = \frac{dr/dt}{d\phi/dt} = \pm \frac{\mu r^2}{L} \sqrt{\frac{2E}{\mu} - \frac{2V}{\mu} - \frac{L^2}{\mu^2 r^2}} \implies d\phi = \pm \frac{L}{\mu r^2} \frac{dr}{\sqrt{\frac{2E}{\mu} - \frac{2V}{\mu} - \frac{L^2}{\mu^2 r^2}}}.$$

Integrating on both sides gives the trajectory of the orbit. We are particularly interested in gravitationally induced orbit, so we should set $V(r) = -k/r$. Note that there is a polynomial in $1/r$ in the r integral. Making a substitution $u = 1/r$, $dr = -r^2 du$,

$$d\phi = \pm \frac{L}{\mu r^2} \frac{dr}{\sqrt{\frac{2E}{\mu} + \frac{2k}{\mu r} - \frac{L^2}{\mu^2 r^2}}} = \mp \frac{L}{\mu} \frac{du}{\sqrt{\frac{2E}{\mu} + \frac{2k}{\mu} u - \frac{L^2}{\mu^2} u^2}}.$$

Define some constants,

$$\tilde{u} = \frac{L}{\mu} u, \quad b = \frac{2E}{\mu}, \quad c = \frac{k}{L}.$$

Then

$$d\phi = \mp \frac{d\tilde{u}}{\sqrt{b + 2c\tilde{u} - \tilde{u}^2}} = \mp \frac{d\tilde{u}}{\sqrt{-(c - \tilde{u})^2 + b + c^2}}.$$

Making another substitution $\sin \alpha = (\tilde{u} - c)/\sqrt{b + c^2}$, $d\tilde{u} = \sqrt{b + c^2} \cos \alpha d\alpha$,

$$d\phi = \mp \frac{\sqrt{b + c^2} \cos \alpha d\alpha}{\sqrt{b + c^2} \sqrt{1 - \sin^2 \alpha}} = \mp d\alpha.$$

Now the RHS is trivial as everything cancels out perfectly, which gives

$$\phi - \phi_0 = \mp \alpha = \mp \arcsin \left(\frac{\tilde{u} - c}{\sqrt{b + c^2}} \right) \implies \tilde{u} = c \mp \sqrt{b + c^2} \sin(\phi - \phi_0).$$

Substitute back all the u , $\tilde{u}$, b , and c into r ,

$$r = \frac{L/\mu}{k/L \mp \sqrt{2E/\mu + (k^2/L^2)} \sin(\phi - \phi_0)} = \frac{L^2/k\mu}{1 + \sqrt{1 + 2EL^2/k^2\mu} \cos \phi}.$$

In the last step, we get rid of the initial phase because we can always choose a ϕ_0 such that the sine becomes a cosine and that there is no $\mp$ sign. Remember we are only interested in the shape. Finally, simplify our expression of r to be

$$\boxed{r = \frac{r_0}{1 + \epsilon \cos \phi}} \quad \text{where} \quad r_0 = \frac{L^2}{k\mu}, \quad \epsilon = \sqrt{1 + \frac{2EL^2}{k^2\mu}}. \quad (3.4)$$

You would recognize this equation as conic sections with eccentricity ϵ . Let's analyze this equation,

1. If $\epsilon = 0$, both E and r is at minimum (and E is negative). The shape is simply a circle with fixed radius $r = r_0$.
2. If $0 < \epsilon < 1$, the shape is an ellipse. The "size" (the semi-major axis) of the ellipse is

$$2a = \frac{r_0}{1 - \epsilon} + \frac{r_0}{1 + \epsilon} = \frac{2r_0}{1 - \epsilon^2} = -\frac{k}{E}.$$

This shows that a smaller energy will have a smaller orbit, and also note that the size of the orbit does not depend on L . L

only affects its eccentricity. We can go back and forth between geometric properties and conserved quantities by using

$$E = -\frac{k}{2a}, \quad L = \sqrt{k\mu a(1 - \epsilon^2)}.$$

Remember that the eccentricity is defined as $\epsilon = \sqrt{1 - b^2/a^2}$, where a and b are semi-major and semi-minor axis respectively.

3. If $\epsilon > 1$, the denominator of (3.4) vanishes at $\phi = \arccos(-1/\epsilon)$. At this angle, $r \rightarrow \infty$, which means the trajectory is unbound. The distance of closest approach is $r_0/(1 + \epsilon)$. Geometrically, the conic sections with eccentricity $\epsilon > 1$ is a hyperbola.
4. $\epsilon = 1$ is the boundary of bound and unbound trajectories. The shape is a parabola.

3.3.2 Kepler's Laws

Kepler postulate three laws that govern the dynamics of planets:

1. Planets move in elliptical orbits, with the sun at one focus.
2. Planetary orbits sweep out equal areas in equal times.
3. The periods squared of planetary orbits are proportional to their semi-major axes cubed.

The first law is already proved in the last section using Lagrangian mechanics. We will prove the second and the third law.

Kepler's Second Law The area element swept by an orbit can be approximated by a triangle with base $r \delta\phi$ and height r ,

$$\delta A = \frac{1}{2} r(r \delta\phi) = \frac{1}{2} r^2 \delta\phi.$$

Taking the limit $\delta t \rightarrow 0$,

$$\frac{dA}{dt} = \frac{1}{2} r^2 \dot{\phi} = \frac{\mu r^2 \dot{\phi}}{2\mu} = \frac{L}{2\mu} = \text{const.}$$

Then if $\Delta t = t_1 - t_2 = t_3 - t_4$, the area swept out by the orbit in Δt is

$$A = \int_{t_1}^{t_2} \left(\frac{dA}{dt} \right) dt = \frac{L\Delta t}{2\mu} = \int_{t_3}^{t_4} \left(\frac{dA}{dt} \right) dt.$$

Therefore, Kepler's second law is equivalent to conservation of angular momentum. Note that it does not have any requirement to the shape of the orbit. It just needs the force to be central so that there is no torque, $\boldsymbol{\tau} = (F\hat{\mathbf{r}}) \times \mathbf{r} = 0$.

Kepler's Third Law According to Kepler's second law, the area swept by the orbit over one period T is

$$\int_0^T \left(\frac{dA}{dt} \right) dt = \frac{LT}{2\mu} = \pi ab,$$

where in the last equality we use the area formula of an ellipse. Using the definition of eccentricity, $\epsilon = \sqrt{1 - b^2/a^2}$, and the angular momentum derived in the last section, $L = \sqrt{k\mu a(1 - \epsilon^2)}$, we can solve for the period,

$$T = \frac{2\mu\pi ab}{L} = \frac{2\mu\pi ab}{\sqrt{k\mu a(1 - \epsilon^2)}} = \frac{2\mu\pi ab}{\sqrt{k\mu a(b^2/a^2)}} = \boxed{\frac{2\pi a^{3/2}}{\sqrt{k/\mu}}}.$$

3.4 Bertrand's Theorem

A closed orbit is an orbit with $r(\phi) = r(\phi + 2n\pi)$ for some integer n and for all ϕ , i.e. the object returns to the same orbit after some period.

Theorem 3.1. **Bertrand's theorem** says that the only central force potentials $V(r)$ for which *all* bound orbits are closed are the following:

1. The inverse-square central force potential $V(r) \propto -1/r$.
2. The harmonic oscillator potential $V(r) \propto r^2$.

To prove this, we first need to analyze the effective potential

$$V_{\text{eff}} = \frac{L^2}{2\mu r^2} + V(r).$$

To have a bound orbit, the graph of V_{eff} vs r must have a minimum. By taking the derivative with respect to r , we can show that the minimum occurs at $r = R$ that satisfies

$$V'_{\text{eff}}(R) = 0 \implies V'(R) = \frac{L^2}{\mu R^3}. \quad (3.5)$$

A *stable* orbit requires

$$V''_{\text{eff}}(R) = \frac{3L^2}{\mu R^4} + V''(R) > 0 \implies V''(R) + \frac{3}{R}V'(R) > 0. \quad (3.6)$$

If the circular orbit is stable, then there must exist some non-circular orbit (or perturbed circular orbit) with $r_{\min}$ and $r_{\max}$ about $r = R$. Define the apsidal angle $\Delta\phi$ between the point on the perturbed orbit at $r_{\min}$ and the point at $r_{\max}$. We also assume that $(R - r_{\min})/R \ll 1$ and $(r_{\max} - R)/R \ll 1$. A closed orbit would require

$$\Delta\phi = 2\pi \frac{m}{n}, \quad m, n \in \mathbb{Z} \quad \forall R.$$

3.4.1 The Perturbation Frequency

The energy equation says that

$$E = \frac{1}{2}\mu\dot{r}^2 + V_{\text{eff}}(r) = \frac{1}{2}\mu\left(\frac{dr}{d\phi}\dot{\phi}\right)^2 + V_{\text{eff}}(r) = \frac{1}{2}\mu\left(\frac{L}{\mu r^2}\right)^2 r'^2 + V_{\text{eff}}(r),$$

where $r' = dr/d\phi$. Again make the substitution $u = 1/r$, $r' = -u'/u^2$,

$$E = \frac{L^2}{2\mu}u'^2 + \frac{L^2}{2\mu}u^2 + V\left(\frac{1}{u}\right) = \frac{L^2}{2\mu}u'^2 + V_{\text{eff}}(u).$$

Consider the small perturbation $u(\phi) = u_0 + \xi(\phi)$, where $u_0 = 1/R$. We can expand $V_{\text{eff}}(u)$ around u_0 ,

$$V_{\text{eff}}(u) = V_{\text{eff}}(u_0) + \xi V'_{\text{eff}}(u_0) + \frac{1}{2}\xi^2 V''_{\text{eff}}(u_0) + \mathcal{O}(\xi^3).$$

The linear term in ξ vanishes because $V'_{\text{eff}}(u_0) = 0$. Substitute $V_{\text{eff}}(u)$ into the energy equation

$$E' \approx \frac{L^2}{2\mu}\xi'^2 + \frac{1}{2}\xi^2 V''_{\text{eff}}(u_0) \quad \text{where} \quad E' = E - V_{\text{eff}}(u_0).$$

This resembles the simple harmonic energy equation

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2.$$

Think $L^2/\mu = m$, $\xi = x$, $\phi = t$, and $V''_{\text{eff}}(u_0) = m\omega^2$. For simplicity, we just write the solution of ξ as $A \cos(\omega\phi)$, and so

$$u(\phi) = u_0 + A \cos(\omega\phi).$$

Now make use of the (3.5) and (3.6),

$$V'(R) = \frac{L^2}{\mu R^3}, \quad V''_{\text{eff}}(R) = V''(R) + \frac{3}{R}V'(R),$$

and the derivatives

$$\begin{aligned} \frac{dV_{\text{eff}}}{du} &= \frac{dV_{\text{eff}}}{dr} \frac{dr}{du} = -r^2 V'(r), \\ \frac{d^2 V_{\text{eff}}}{du^2} &= \frac{d}{du} \left(\frac{dV_{\text{eff}}}{du} \right) = \frac{dr}{du} \frac{d}{dr} [-r^2 V'(r)] = r^2 [2r V'_{\text{eff}}(r) + r^2 V''_{\text{eff}}(r)] \implies \frac{d^2 V_{\text{eff}}(R)}{du^2} = R^4 V''_{\text{eff}}(R). \end{aligned}$$

The frequency ω can be written as

$$\omega = \sqrt{\frac{V''_{\text{eff}}(u_0)}{m}} = \sqrt{\frac{\mu V''_{\text{eff}}(u_0)}{L^2}} = \sqrt{\frac{R^4 [V''(R) + 3V'(R)/R]}{R^3 V'(R)}} = \sqrt{\frac{3V'(R) + R V''(R)}{V'(R)}}.$$

By the definition of apsidal angle $\Delta\phi$, u changes from $u_{\text{max}} = u_0 + A$ and $u_{\text{min}} = u_0 - A$ within $\Delta\phi$. This means $\omega\Delta\phi = \pi$, or $\Delta\phi = \pi/\omega$.

3.4.2 The Power of the Potential

In general, any physical potential can be expanded in power series of r plus a logarithmic term,

$$V(r) = \sum_{n=-\infty}^{\infty} \frac{a_n}{r^n} + a \ln r.$$

We want to have an apsidal angle independent of r (or else the apsidal angle will be constantly changing throughout the orbit). Let $\eta = V'(r)$, and define the constant β such that

$$\omega = \sqrt{3 + R \frac{\eta'}{\eta}} \equiv \sqrt{\beta}, \quad \beta > 0.$$

Rearranging some terms, we get a simple differential equation,

$$\frac{R\eta'}{\eta} = \beta - 3.$$

Its solution is

$$\eta(r) = kr^{\beta-3} \implies V(r) = \frac{k}{\beta-2} r^{\beta-2} = -\frac{k}{\alpha} r^{-\alpha},$$

where $\alpha = 2 - \beta$. The domain of β also restricts $\alpha < 2$ and $\alpha \neq 0$, unless $V(r) \propto \ln r$. Substitute $\beta = 2 - \alpha$ into the apsidal angle, we must have

$$\Delta\phi = \frac{\pi}{\sqrt{2-\alpha}}$$

for it to be independent of r . For $\sqrt{2-\alpha}$ to be rational, there are infinite solutions for α . However, this analysis is based on small perturbations. We shall see that only when $\alpha = 1$ and $\alpha = -2$ the orbit is exactly closed, no matter the extent of deviation from the

circular orbit. Now let's divide the potential into two cases: one being $\alpha < 0$ and the other $0 < \alpha < 2$. The $\alpha = 0$ case does not form a closed orbit.

3.4.3 The Harmonic Oscillator

When $\alpha < 0$,

$$\lim_{r \rightarrow \infty} V(r) = \lim_{r \rightarrow \infty} \left(-\frac{k}{\alpha} r^{-\alpha} \right) = \infty.$$

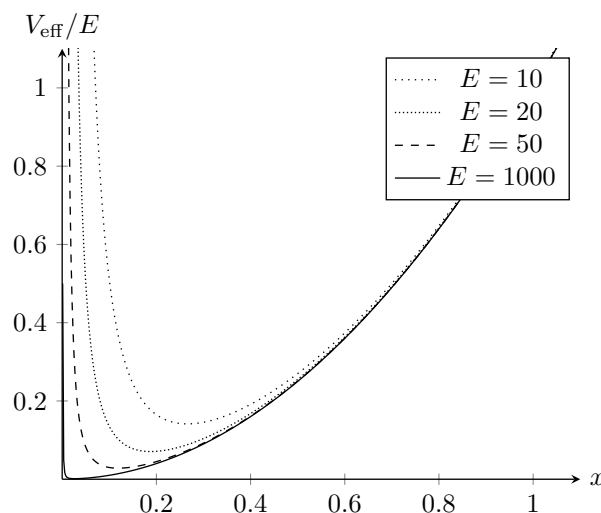
We consider the extreme case where $E \rightarrow \infty$. (This is a bound orbit because $V(\infty) = \infty$). If the potential $V(r) = -kr^{-\alpha}/\alpha$ does not produce a closed orbit in this case, then the corresponding α is ruled out of Bertrand's theorem. At $u_{\max}$ (or $r_{\min}$),

$$E = \frac{L^2}{2\mu} u'^2 + \frac{L^2}{2\mu} u^2 + V\left(\frac{1}{u}\right) = \frac{L^2}{2\mu} u'^2 + \frac{L^2}{2\mu} u_{\max}^2 - \frac{k}{\alpha} u_{\max}^\alpha \simeq \frac{L^2}{2\mu} u_{\max}^2.$$

The last term is small comparing to the second because α is a negative number, especially as $E \rightarrow \infty$. The first term is literally zero because $u' = 0$ at $u_{\max}$, otherwise it wouldn't be a maximum. With this result, define $x = u/u_{\max}$. The effective potential becomes

$$V_{\text{eff}}(u) = \frac{L^2}{2\mu} u'^2 - \frac{k}{\alpha} u^\alpha = \frac{L^2}{2\mu} x^2 u_{\max}^2 - \frac{k}{\alpha} x^\alpha u_{\max}^\alpha = x^2 E - \frac{k}{\alpha} \left(\frac{L^2}{2\mu} \right)^{-\alpha/2} E^{\alpha/2} x^\alpha.$$

As $E \rightarrow \infty$, the effective potential looks like a half-harmonic oscillator—a harmonic oscillator potential but with a wall near $x = 0$:



This happens because the second term is negligible except near $x = 0$. We can plug V_{eff} into the energy equation,

$$E = \frac{L^2}{2\mu} u'^2 + V_{\text{eff}} = \frac{L^2}{2\mu} u'^2 + x^2 E = \frac{L^2}{2\mu} u'^2 + \left(\frac{u}{u_{\max}} \right)^2 \left(\frac{L^2}{2\mu} u_{\max}^2 \right) = \frac{L^2}{2\mu} u'^2 + \frac{L^2}{2\mu} u^2.$$

This is again the energy equation of a simple harmonic oscillator, this time with $\omega = 1$. But because of the half harmonic oscillator potential, it only takes one quarter of the harmonic cycle for the object to travel from $u_{\max}$ to $u_{\min} = 0$. Thus,

$$\omega \Delta\phi = \frac{\pi}{2} \quad \implies \quad \Delta\phi = \frac{\pi}{2}.$$

Compare this with $\Delta\phi = \pi/\sqrt{2-\alpha}$, we have $\alpha = -2$. The potential looks like

$$V(r) = -\frac{k}{\alpha} r^{-\alpha} = \frac{k}{2} r^2.$$

3.4.4 The Inverse-Square Force Potential

When $0 < \alpha < 2$, all bound orbits have $E < 0$. We take the loosely bound limit $E \rightarrow 0$. The energy equation is then

$$\frac{L^2}{2\mu} u'^2 + \frac{L^2}{2\mu} u^2 - \frac{k}{\alpha} u^\alpha = 0.$$

Dividing both sides by u^α and move k/α to the RHS,

$$\frac{L^2}{2\mu} u^{-\alpha} u'^2 + \frac{L^2}{2\mu} u^{2-\alpha} = \frac{k}{\alpha}.$$

Define $y^2 = u^{2-\alpha}$, $2 dy/y = (2 - \alpha) du/u$, and so

$$\frac{L^2}{2\mu} u^{2-\alpha} \left(\frac{du}{u d\phi} \right)^2 + \frac{L^2}{2\mu} u^{2-\alpha} = \frac{k}{\alpha} \implies \frac{L^2}{2\mu} \left(\frac{2}{2-\alpha} \right)^2 \left(\frac{dy}{d\phi} \right)^2 + \frac{L^2}{2\mu} y^2 = \frac{k}{\alpha}.$$

This is again a simple harmonic oscillator equation, with solution

$$y = A \cos(\omega\phi) \quad \text{where} \quad \omega = \frac{2-\alpha}{2}.$$

The bounds are $r_{\max} = \infty$ ($y_{\min} = 0$) and $r_{\min}$ ($y_{\max}$). To oscillate from $y_{\max}$ to $y_{\min} = 0$, it again only takes one quarter of a harmonic cycle, so

$$\omega \Delta\phi = \frac{\pi}{2} \implies \Delta\phi = \frac{\pi}{2-\alpha}.$$

Compare it with $\Delta\phi = \pi/\sqrt{2-\alpha}$, we find that $\alpha = 1$. The potential looks like

$$V(r) = -\frac{k}{\alpha} r^{-\alpha} = -\frac{k}{r}.$$

In conclusion, only r^2 and $-1/r$ potential satisfies both perturbation analysis and asymptotic analysis. Indeed, their bound orbits can be determined easily: both are ellipses. (It is extremely easy to show that r^2 potential has elliptical orbit because its potential is also separable, $r^2 = x^2 + y^2 + z^2$.) Their difference is that $-1/r$ potential has the source at one of the foci of the ellipse, while r^2 potential has the source at the center. This is the reason why $\Delta\phi = \pi$ for $-1/r$ potential but $\Delta\phi = \pi/2$ for r^2 potential, even though their orbits are both ellipses.

3.5 A Hidden Symmetry: the Laplace-Runge-Lenz Vector

Back to the two-body central force problem with $V = -k/r$, its Lagrangian is

$$\mathcal{L} = \frac{1}{2}\mu(\dot{x}^2 + \dot{y}^2) + \frac{k}{\sqrt{x^2 + y^2}},$$

expressed in Cartesian coordinates in the CM frame. One can use Lagrange equations to find the equations of motion for this system:

$$\ddot{x} = -\frac{kx}{\mu(x^2 + y^2)^{3/2}}, \quad \ddot{y} = -\frac{ky}{\mu(x^2 + y^2)^{3/2}}.$$

Consider this “strange” transformation

$$\xi_x = -\frac{\beta}{2}\mu y\dot{y}, \quad \xi_y = \frac{\beta}{2}\mu(2x\dot{y} - y\dot{x}),$$

where β is some (dimensionful) constant. The change in the Lagrangian under this transformation is

$$\begin{aligned}\delta\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial q_i}\epsilon\xi_i + \frac{\partial\mathcal{L}}{\partial\dot{q}_i}\epsilon\dot{\xi}_i = \epsilon\left(\frac{\partial\mathcal{L}}{\partial x}\xi_x + \frac{\partial\mathcal{L}}{\partial y}\xi_y + \frac{\partial\mathcal{L}}{\partial\dot{x}}\dot{\xi}_x + \frac{\partial\mathcal{L}}{\partial\dot{y}}\dot{\xi}_y\right) \\ &= \frac{\beta}{2}\epsilon\mu\left[\frac{kxy\dot{y}}{(x^2+y^2)^{3/2}} - \mu\dot{x}\dot{y}^2 - \mu\dot{x}y\ddot{y} - \frac{2kxy\dot{y}}{(x^2+y^2)^{3/2}} + \frac{k\dot{x}y^2}{(x^2+y^2)^{3/2}} + 2\mu\dot{x}\dot{y}^2 + 2\mu x\dot{y}\ddot{y} - \mu\dot{x}\dot{y}^2 - \mu\ddot{x}y\dot{y}\right] \\ &= \frac{\beta}{2}\epsilon\mu^2\left[\frac{ky(\dot{x}y - x\dot{y})}{\mu(x^2+y^2)^{3/2}} - \dot{x}y\ddot{y} + 2x\dot{y}\ddot{y} - \ddot{x}y\dot{y}\right].\end{aligned}$$

Now substitute the equations of motion for $\ddot{x}$ and $\ddot{y}$,

$$\delta\mathcal{L} = \frac{\beta}{2}\epsilon\mu^2\left[\frac{ky(\dot{x}y - x\dot{y})}{\mu(x^2+y^2)^{3/2}} + \dot{x}y\frac{ky}{\mu(x^2+y^2)^{3/2}} - 2x\dot{y}\frac{ky}{\mu(x^2+y^2)^{3/2}} + \frac{kx}{\mu(x^2+y^2)^{3/2}}y\dot{y}\right] = \beta\epsilon\mu k\frac{y(\dot{x}y - x\dot{y})}{(x^2+y^2)^{3/2}}.$$

One can verify that the final expression is a total derivative with respect to time,

$$\delta\mathcal{L} = \beta\epsilon\mu k\frac{d}{dt}\left(\frac{x}{\sqrt{x^2+y^2}}\right).$$

Since the change in the Lagrangian is a total time derivative of some quantity $K = x/\sqrt{x^2+y^2}$, it is a symmetry of the physical trajectory, or an **on-shell** symmetry. By Noether's theorem, there exists a conserved quantity corresponding to this symmetry. We denote this quantity by A_x ,

$$\begin{aligned}A_x &\propto p_i\xi_i - K = \frac{\partial\mathcal{L}}{\partial\dot{x}}\xi_x + \frac{\partial\mathcal{L}}{\partial\dot{y}}\xi_y - K \\ &\propto \mu\dot{x}\left(-\frac{\beta}{2}\mu y\dot{y}\right) + \mu\dot{y}\left[\frac{\beta}{2}\mu(2x\dot{y} - y\dot{x})\right] - \beta\mu k\frac{x}{\sqrt{x^2+y^2}} \propto \beta\mu^2(x\dot{y}^2 - y\dot{x}\dot{y}) - \beta\mu k\frac{x}{\sqrt{x^2+y^2}}.\end{aligned}$$

We can choose the proportionality constant to be β^{-1} to cancel the β . Since the system has $x \leftrightarrow y$ symmetry, exchanging x and y in the reasoning above gives another conserved quantity,

$$A_y = \mu^2(\dot{x}^2y - x\dot{x}\dot{y}) - \mu k\frac{y}{\sqrt{x^2+y^2}}.$$

There is no A_z , so we can construct a vector along the xy -plane,

$$\begin{aligned}\mathbf{A} &= A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} \\ &= \mu^2(x\dot{y}^2 - y\dot{x}\dot{y})\hat{\mathbf{x}} + \mu^2(\dot{x}^2y - x\dot{x}\dot{y})\hat{\mathbf{y}} - \mu k\frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}}}{\sqrt{x^2+y^2}} \\ &= \mu^2(x\dot{y} - y\dot{x})(-\dot{x}\hat{\mathbf{x}} + \dot{y}\hat{\mathbf{y}}) - \mu k\frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}}}{\sqrt{x^2+y^2}} \\ &= \mu^2(\dot{x}\hat{\mathbf{x}} + \dot{y}\hat{\mathbf{y}}) \times (x\dot{y} - y\dot{x})\hat{\mathbf{z}} - \mu k\frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}}}{\sqrt{x^2+y^2}}\end{aligned}$$

The first bracket is just the velocity vector, while the second bracket can be written in terms of a cross product,

$$\mathbf{A} = \mu^2\mathbf{v} \times \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ x & y & 0 \\ \dot{x} & \dot{y} & 0 \end{vmatrix} - \mu k\frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}}}{\sqrt{x^2+y^2}} = \mu\mathbf{v} \times (\mathbf{r} \times \mu\mathbf{v}) - \mu k\hat{\mathbf{r}}$$

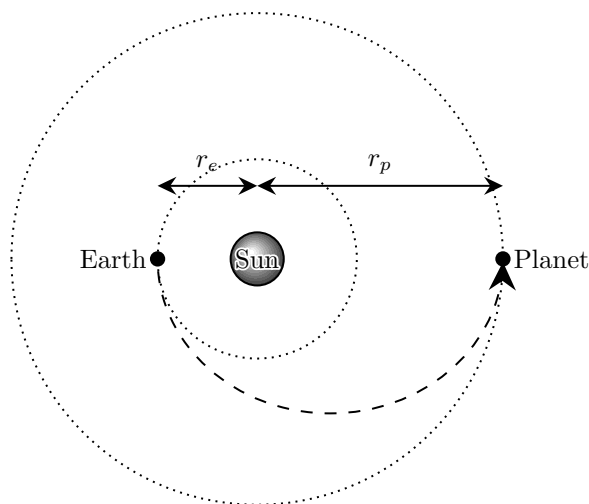
The final expression is called the **Laplace-Runge-Lenz vector**:

$$\boxed{\mathbf{A} = \mathbf{p} \times \mathbf{L} - \mu k\hat{\mathbf{r}}} \quad (3.7)$$

You can sketch the Laplace-Runge-Lenz vector along the elliptical trajectory in the xy -plane with the vectors $\mathbf{r}$, $\mathbf{v}$, and $\mathbf{L}$. The result is that $\mathbf{A}$ always point parallel to the major-axis of the ellipse. The conservation of $\mathbf{A}$ is the reason that the elliptical orbits don't precess, i.e. they remain in a closed orbit. If there are more than two objects interacting, $\mathbf{A}$ is no longer conserved. This is also the reason why (only) the hydrogen atom has an analytical solution in terms of ladder operators in quantum mechanics. In fact, Wolfgang Pauli derived the spectrum of hydrogen using $\mathbf{A}$ *before the development of the Schrödinger equation!*

3.6 Transfer of Orbits

The most common way to transfer between orbits is via a **Hohmann transfer orbit** because it requires the least fuel. The idea is to first speed up the spacecraft to gain energy. The spacecraft will then follow an elliptical orbit until it reaches the orbit of the designated planet. Finally, it modify its speed to reach the speed of the planet.



Assume the Earth and the planet follow circular orbits with radius r_e and r_p , respectively. The semi-major axis of the transfer orbit is then $a = (r_e + r_p)/2$. The energy is given by

$$E = -\frac{k}{2a} = -\frac{GMm}{2a},$$

where M is the mass of the sun and m is the mass of the spacecraft. At the surface of the Earth, the speed of the spacecraft should satisfy

$$\frac{1}{2}mv^2 - \frac{GMm}{r_e} = -\frac{GMm}{2a} \implies v = \sqrt{2GM \left(\frac{1}{r_e} + \frac{1}{r_e + r_p} \right)} = \sqrt{2GM \frac{r_p}{r_e(r_e + r_p)}}.$$

At the planet,

$$\frac{1}{2}mv^2 - \frac{GMm}{r_p} = -\frac{GMm}{2a} \implies v = \sqrt{2GM \left(\frac{1}{r_p} - \frac{1}{r_e + r_p} \right)} = \sqrt{2GM \frac{r_e}{r_p(r_e + r_p)}}.$$

If we ignore the gravity of along the trip, then the change in speed at the Earth and at the planet would be

$$\Delta v_e = \sqrt{2GM \frac{r_p}{r_e(r_e + r_p)}} - \sqrt{\frac{GM}{r_e}}, \quad \Delta v_p = \sqrt{\frac{GM}{r_p}} - \sqrt{2GM \frac{r_e}{r_p(r_e + r_p)}}.$$

4 NON-INERTIAL FRAMES

In almost all scenarios, we emphasize setting up an inertial frame for physics problems. Inertial frames are ones in which an isolated object moves with constant velocity, consistent with Newton's first law and second law. In a non-inertial frame, even if there is no force acting on an object, it will still accelerate. We may wonder how to formulate “accelerations” in non-inertial frames. The general solution is provided by Lagrangian mechanics. In a non-inertial frame, the object is still following the physical trajectory in which the action is stationary. The only thing has changed is the coordinates used to describe the trajectory. Previously, we chose our generalized coordinates such that they make mathematics easier. For non-inertial frames, we will choose the generalized coordinates such that they represent the motion of the system as it appears to a non-inertial observer.

4.1 A Uniformly Accelerating Frame

Let $\mathbf{a}_0$ be the acceleration of the non-inertial frame S relative to some inertial frame S' . Then the change of coordinates between S and S' is

$$x' = x + \frac{1}{2}a_{0,x}t^2, \quad y' = y + \frac{1}{2}a_{0,y}t^2, \quad z' = z + \frac{1}{2}a_{0,z}t^2,$$

where the primed coordinates denote those in the inertial frame and unprimed coordinates are for the non-inertial frame. Also, we choose that at $t = 0$ the origins of the two frames coincide with no relative velocity. For a free particle in S' , the Lagrangian is

$$\mathcal{L} = \frac{1}{2}m(\dot{x}'^2 + \dot{y}'^2 + \dot{z}'^2).$$

Making a transformation into S , the Lagrangian in terms of non-inertial coordinates is

$$\mathcal{L} = \frac{1}{2}m[(\dot{x} + a_{0,x}t)^2 + (\dot{y} + a_{0,y}t)^2 + (\dot{z} + a_{0,z}t)^2] = \frac{1}{2}m(\mathbf{v} + \mathbf{a}_0t)^2.$$

Applying Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = 0 \quad \implies \quad m\ddot{x} + ma_{0,x} = 0.$$

We can do the same for y and z . Writing in vector form, we have

$$m\mathbf{a} - m\mathbf{a}_0 = 0 \quad \implies \quad \mathbf{a} = -\mathbf{a}_0.$$

Here $\mathbf{a}$ is the acceleration of the particle in the S' frame. It means that in the accelerating frame, an observer will find that a free particle is accelerating backwards with magnitude a_0 . Think about sitting in a car: when the car accelerate forward relative to the ground frame, you experience a “force” that pulls you backward. Such effects are known as **pseudoforces**. Since there is no actual physical interactions making the particle accelerate, we won't call it an actual force.

One can also interpret this pseudoforce as an effective gravity $\mathbf{g}_{\text{eff}} = -\mathbf{a}_0$ (which will be super important in Einstein's general relativity). First, let us consider some actual gravity with gravitational acceleration $\mathbf{g}$ in both S and S' . The corresponding potential is $V = -m\mathbf{g} \cdot \mathbf{r}$. The Lagrangian in S' is

$$\mathcal{L} = \frac{1}{2}m(\mathbf{v} + \mathbf{a}_0t)^2 + m\mathbf{g} \cdot \mathbf{r}.$$

By Lagrange equations, we will find that

$$m\mathbf{g} = m\mathbf{a} + m\mathbf{a}_0 \implies \mathbf{a} = \mathbf{g} - \mathbf{a}_0.$$

The particle accelerates as though there is a uniform gravitational field $\mathbf{g} - \mathbf{a}_0$ instead of $\mathbf{g}$. For example, a pendulum hanged in a car accelerating forward will be tilted backwards instead of hanging straight down. Interestingly, with the effective gravity interpretation, one can just let $V_{\text{eff}} = -m\mathbf{g} \cdot \mathbf{r} + m\mathbf{a}_0 \cdot \mathbf{r}$, and write the Lagrangian,

$$\mathcal{L}' = \frac{1}{2}m\mathbf{v}^2 - m\mathbf{a}_0 \cdot \mathbf{r} + m\mathbf{g} \cdot \mathbf{r}.$$

It produces exactly the same equation of motion. This is because

$$\mathcal{L} - \mathcal{L}' = m\mathbf{v} \cdot \mathbf{a}_0 t + \frac{1}{2}ma_0 t^2 + m\mathbf{a}_0 \cdot \mathbf{r} = \frac{d}{dt} \left(m\mathbf{r} \cdot \mathbf{a}_0 t + \frac{1}{6}ma_0 t^3 \right).$$

The difference in Lagrangians is a total time derivative of some function which does not depends on $\mathbf{v}$. By Theorem 2.1, we know that they product the same equations of motion.

4.2 A Rotating Frame

In general, a rotating frame means the frame of reference of an observer who is rotating around some fixed central point. We can choose the origin of the rotating frame to be the same as the origin of the inertial frame. As before, we denote primed coordinates for the inertial frame and unprimed coordinates for the rotating frame. The position vectors of some object in the two frames are

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}} = x'\hat{\mathbf{x}}' + y'\hat{\mathbf{y}}' + z'\hat{\mathbf{z}}' = \mathbf{r}'.$$

The position vectors are the same because S' and S have the same origin. It is just the coordinates and components are different. Differentiating the position vector with respect to time,

$$\frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\hat{\mathbf{x}} + x\frac{d\hat{\mathbf{x}}}{dt} + \frac{dy}{dt}\hat{\mathbf{y}} + y\frac{d\hat{\mathbf{y}}}{dt} + \frac{dz}{dt}\hat{\mathbf{z}} + z\frac{d\hat{\mathbf{z}}}{dt}.$$

We will define the velocity in the rotating frame to be

$$\mathbf{v}_{\text{rot}} \equiv \dot{x}\hat{\mathbf{x}} + \dot{y}\hat{\mathbf{y}} + \dot{z}\hat{\mathbf{z}}.$$

The physical meaning of $\mathbf{v}_{\text{rot}}$ is that it is the velocity of a particle observed in the rotating frame. For example, if a particle is moving away along the x -axis, then the observer in the rotating frame will just perceive a velocity of $\dot{x}\hat{\mathbf{x}}$. However, for the inertial observer, the x -axis is also changing in time, so there will be extra velocities to consider, namely $x d\hat{\mathbf{x}}/dt$.

We should write the basis vector $\hat{\mathbf{x}}$ in the primed coordinates to see what happens. Let the angular velocity of the rotating frame be ω_z (not necessarily uniform). Suppose at some time t , the rotating basis vector can be expressed as $\hat{\mathbf{x}}(t) = \hat{\mathbf{x}}' \cos \phi + \hat{\mathbf{y}}' \sin \phi$. After some time dt , the new $\hat{\mathbf{x}}$ is

$$\begin{aligned} \hat{\mathbf{x}}(t + dt) &= \hat{\mathbf{x}}' \cos(\phi + \omega_z dt) + \hat{\mathbf{y}}' \sin(\phi + \omega_z dt) = \hat{\mathbf{x}}' \\ &= \hat{\mathbf{x}}' [\cos \phi \cos(\omega_z dt) - \sin \phi \sin(\omega_z dt)] + \hat{\mathbf{y}}' [\sin \phi \cos(\omega_z dt) + \cos \phi \sin(\omega_z dt)] \\ &= \hat{\mathbf{x}}' [\cos \phi - (\omega_z dt) \sin \phi] + \hat{\mathbf{y}}' [\sin \phi + (\omega_z dt) \cos \phi] \\ &= (\hat{\mathbf{x}}' \cos \phi + \hat{\mathbf{y}}' \sin \phi) + \omega_z dt (\hat{\mathbf{y}}' \cos \phi - \hat{\mathbf{x}}' \sin \phi) \\ &= \hat{\mathbf{x}}(t) + \hat{\mathbf{y}}(t)(\omega_z dt). \end{aligned}$$

This gives $d\hat{\mathbf{x}}(t) = \omega_z \hat{\mathbf{y}} dt$, or $d\hat{\mathbf{x}}/dt = \omega_z \hat{\mathbf{y}}$. We can use the same method to find derivatives of $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$, and for general angular velocity vector $\boldsymbol{\omega}$. The result is

$$\frac{d\hat{\mathbf{n}}}{dt} = \boldsymbol{\omega} \times \hat{\mathbf{n}}, \quad \text{where } n = x, y, z.$$

Then we can write

$$\mathbf{v}' = \frac{d\mathbf{r}}{dt} = \dot{x}\hat{\mathbf{x}} + \dot{y}\hat{\mathbf{y}} + \dot{z}\hat{\mathbf{z}} + x(\boldsymbol{\omega} \times \hat{\mathbf{x}}) + y(\boldsymbol{\omega} \times \hat{\mathbf{y}}) + z(\boldsymbol{\omega} \times \hat{\mathbf{z}}) = \mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{r}.$$

The Lagrangian for a free particle is

$$\begin{aligned} \mathcal{L} &= \frac{1}{2}m\mathbf{v}'^2 = \frac{1}{2}m(\mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{r})^2 \\ &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + \frac{1}{2}m(\boldsymbol{\omega} \times \mathbf{r})^2 + m(\dot{x}\hat{\mathbf{x}} + \dot{y}\hat{\mathbf{y}} + \dot{z}\hat{\mathbf{z}}) \cdot (\boldsymbol{\omega} \times \mathbf{r}) \\ &= \frac{1}{2}m[\dot{x}^2 + \dot{y}^2 + \dot{z}^2 + (\omega_y z - \omega_z y)^2 + (\omega_z x - \omega_x z)^2 + (\omega_x y - \omega_y x)^2 \\ &\quad + 2\dot{x}(\omega_y z - \omega_z y) + 2\dot{y}(\omega_z x - \omega_x z) + 2\dot{z}(\omega_x y - \omega_y x)]. \end{aligned}$$

In this form, the Lagrangian does not depend on unit vectors, so we don't need to worry about its time-derivative in Lagrange equations. Now we will find the equations of motion:

$$\frac{\partial \mathcal{L}}{\partial x} = m\omega_z(\omega_z x - \omega_x z) + m\omega_y(\omega_y x - \omega_x y) + m\dot{y}\omega_z - m\dot{z}\omega_y,$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{d}{dt} [m\dot{x} + m(\omega_y z - \omega_z y)] = m\ddot{x} + \omega_y \dot{z} - \omega_z \dot{y} + \frac{d\omega_y}{dt} z - \frac{d\omega_z}{dt} y.$$

Plug them into the Lagrange equation,

$$\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = 0 \implies \ddot{x} = 2\omega_z \dot{y} - 2\omega_y \dot{z} + (\omega_y^2 + \omega_z^2)x - \omega_x(y + z) - \frac{d\omega_y}{dt} z + \frac{d\omega_z}{dt} y.$$

The same holds for y and z . Writing the final result in compact vector notation, we have

$$\mathbf{a}_{\text{rot}} = -\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) - 2\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} - \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r}, \quad \mathbf{a}_{\text{rot}} \equiv \ddot{x}\hat{\mathbf{x}} + \ddot{y}\hat{\mathbf{y}} + \ddot{z}\hat{\mathbf{z}}.$$

The quantity $\mathbf{a}_{\text{rot}}$ is the acceleration in the rotating frame. This is what we want to analyze the pseudoforces. There are three pseudoforces arising from a rotating frame:

1. The **centrifugal force**,

$$-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}).$$

It is perpendicular to the axis of rotation, $\hat{\boldsymbol{\omega}}$, and points away from the axis.

2. The **Coriolis force**,

$$-2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}.$$

It deflects a moving particle in a direction perpendicular to its motion.

3. The **Euler pseudoforce**,

$$-m \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r}.$$

It causes the particle to deflect if the rotation of the frame is changing.

5 COUPLED OSCILLATORS

In this chapter, we are interested in **coupled oscillators** with multiple masses connected by multiple springs. The general idea can be extended to atoms, molecules, circuits, elastic solid, etc.

5.1 Two Masses Connected by Three Springs

One of the simplest coupled oscillators consists of two masses m_1 and m_2 , and three springs, lining up in one dimension. Each mass is connected to a (unmovable) wall with a spring with spring constant k . Meanwhile, there is an additional spring with spring constant k' connecting the two masses. It is this spring that couples the two masses and makes the motion more complicated.

Let x_1 and x_2 denotes the deviation from individual's equilibrium position. For now we assume that $m_1 = m_2 = m$ for the simplest case. The Lagrangian of the system is

$$\mathcal{L} = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 - \frac{1}{2}k'(x_1 - x_2)^2.$$

We can easily get the equations of motion by applying Lagrange equations:

$$m\ddot{x}_1 = -kx_1 - k'(x_1 - x_2), \quad m\ddot{x}_2 = -kx_2 + k'(x_1 - x_2).$$

The equations are linear but coupled, i.e. x_1 equation depends on x_2 , and vice versa. This is usually what coupled oscillators would produce. The most general way is to decouple the equations by some coordinate transformation and solve for the new equations. For this problem, note that x_1 and x_2 are symmetric, so it is better to add and subtract the two equations,

$$m(\ddot{x}_1 + \ddot{x}_2) = -k(x_1 + x_2), \quad m(\ddot{x}_1 - \ddot{x}_2) = -k(x_1 - x_2) - 2k'(x_1 - x_2).$$

Let $\xi_+ \equiv x_1 + x_2$ and $\xi_- \equiv x_1 - x_2$,

$$m\ddot{\xi}_+ = -k\xi_+, \quad m\ddot{\xi}_- = -(k + 2k')\xi_-.$$

Each equation now is an equation of simple harmonic motion. The solutions are

$$\begin{cases} \xi_+ = C_+ e^{i\omega_+ t} + C_+^* e^{-i\omega_+ t} = A_+ \cos(\omega_+ t + \phi_+), & \omega_+ = \sqrt{\frac{k}{m}}, \\ \xi_- = C_- e^{i\omega_- t} + C_-^* e^{-i\omega_- t} = A_- \cos(\omega_- t + \phi_-), & \omega_- = \sqrt{\frac{k + 2k'}{m}}. \end{cases}$$

The constants A_+ , A_- , ϕ_+ and ϕ_- will be fixed by initial conditions. Inverting the transformations gives

$$\begin{cases} x_1 = \frac{1}{2}(\xi_+ + \xi_-) = \frac{A_+}{2} \cos(\omega_+ t + \phi_+) + \frac{A_-}{2} \cos(\omega_- t + \phi_-), \\ x_2 = \frac{1}{2}(\xi_+ - \xi_-) = \frac{A_+}{2} \cos(\omega_+ t + \phi_+) - \frac{A_-}{2} \cos(\omega_- t + \phi_-). \end{cases}$$

Now let's analyze some special cases. First, let $A_2 = 0$. Both masses will oscillate at the same frequency, amplitude, and phase. The spring connecting m_1 and m_2 is trivial, since the distance between m_1 and m_2 never changes. You may interpret this as a block of mass $2m$ oscillating between the walls. The second special case is $A_1 = 0$. The two masses oscillate at the same frequency

and amplitude, but opposite phase. That is, they always move opposite to each other, like vibration in a diatomic molecule. These two cases are known as **normal modes**: everything oscillates simple-harmonically at the same frequency. The most general initial conditions will produce complicated oscillations for both masses—they will have different amplitudes and phases at any time. It is also hard to incorporate the idea of frequency because x_1 and x_2 are some linear combination of two simple harmonic motions.

5.2 More Complicated Coupled Oscillators

5.2.1 The Spring Constant Matrix and the Mass Matrix

In general, the system of coupled oscillators do not have objects with the same mass, or springs with the same spring constants. It can even have more than two objects that are inter-connected. It will be hard to come up with a coordinate transformation that decouples equations of motion. We need a more systematic tool. Consider again the Lagrangian in the last example,

$$\mathcal{L} = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 - \frac{1}{2}k'(x_1 - x_2)^2.$$

How will this Lagrangian be modified if we are looking for the most general system of coupled oscillators? As mentioned above, the most general system consists of many oscillators, each oscillating in more than one dimension. We will use generalized coordinates q_i , where $i = 1, 2, 3, \dots, N$ for N degrees of freedom. One would expect some quadratic terms like $\frac{1}{2}q_i^2$ or $\frac{1}{2}kq_iq_j$ that characterize the potential energy. We can write

$$\mathcal{L} \supset -\frac{1}{2}q_i K_{ij} q_j,$$

where K_{ij} is some “spring constant matrix” that provides the coefficients for the quadratic terms. Note that Einstein summation is used,

$$q_i K_{ij} q_j = K_{ij} q^i q^j = \sum_{i,j} q_i K_{ij} q_j.$$

In vector and matrix notation, this is $q_i K_{ij} q_j = \mathbf{q}^T \hat{\mathbf{K}} \mathbf{q}$. For example, in the above two masses three springs case, the spring constant matrix can be written as

$$K_{ij} = \begin{bmatrix} k + k' & -k' \\ -k' & k + k' \end{bmatrix}_{ij}.$$

You can write the spring constant matrix in other forms but still produce the same potential. By convention, we always make K_{ij} symmetric, $K_{ij} = K_{ji}$. Similarly, the kinetic energy may also contains cross terms based on our choice of generalized coordinates. For example, one mass is hung from the ceiling by a spring, and a second mass is hung from the first mass by another spring. If q_1 and q_2 are their coordinates relative to the equilibrium position, the position of the second mass is $q_1 + q_2$, producing a cross term, $(\dot{q}_1 + \dot{q}_2)^2 = \dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2$. Therefore, we will have a (symmetric) “mass matrix” such that

$$\mathcal{L} \supset \frac{1}{2}\dot{q}_i M_{ij} \dot{q}_j.$$

5.2.2 Why Only Quadratic Terms?

Not all potentials are quadratic in coordinates, but coupled oscillators still appears in many other physical systems. This is because in many cases we only care about some object’s motion around an equilibrium point. A well-behaved potential can always be expressed

as a Taylor series around an equilibrium point. For a potential $V(q_1, q_2, \dots, q_N)$, its multivariable Taylor expansion is

$$V(q_1, q_2, \dots, q_N) = V_{\text{eq.}} + \left. \frac{\partial V}{\partial q_i} \right|_{\text{eq.}} q_i + \frac{1}{2} \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{\text{eq.}} q_i q_j + \mathcal{O}(q_i^3).$$

Each repeated index indicates a sum. If we only care about the motion near the equilibrium, then by the definition of equilibrium, the first-derivatives vanish. As usual, we can always shift the potential by some constants so that $V = 0$ at equilibrium. For small oscillations, higher order terms $\mathcal{O}(q_i^3)$ are safely ignored. Then we arrive at

$$V \approx \frac{1}{2} \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{\text{eq.}} q_i q_j.$$

In even more general cases, the kinetic energy term may also depend on the coordinates,

$$T = f_{ij}(q_1, q_2, \dots, q_N) \dot{q}_i \dot{q}_j.$$

For small displacements,

$$T = \dot{q}_i \dot{q}_j f_{ij} \Big|_{\text{eq.}} + \dot{q}_i \dot{q}_j \left. \frac{\partial f_{ij}}{\partial q_k} \right|_{\text{eq.}} q_k + \mathcal{O}(q_k^2).$$

If q_k and the “displacement of masses” are small, we can make the approximation

$$T \approx \dot{q}_i \dot{q}_j f_{ij} \Big|_{\text{eq.}} \implies M_{ij} = 2f_{ij} \Big|_{\text{eq.}}.$$

5.3 General Coupled Oscillators

Now we focus on the system with small oscillations around some equilibrium point. The Lagrangian is

$$\mathcal{L} = \frac{1}{2} \dot{q}_i M_{ij} \dot{q}_j - \frac{1}{2} q_i K_{ij} q_j. \quad (5.1)$$

The problem is to determine $q_i(t)$. To do this, we need to know that

$$\frac{\partial q_i}{\partial q_k} = \delta_{ik} \quad \text{and} \quad \frac{\partial \dot{q}_i}{\partial \dot{q}_k} = \delta_{ik}.$$

The position-derivative of the Lagrangian is then

$$\frac{\partial \mathcal{L}}{\partial q_k} = -\frac{1}{2} \frac{\partial q_i}{\partial q_k} K_{ij} q_j - \frac{1}{2} q_i K_{ij} \frac{\partial q_j}{\partial q_k} = -\frac{1}{2} \delta_{ik} K_{ij} q_j - \frac{1}{2} q_i K_{ij} \delta_{jk} = -\frac{1}{2} K_{kj} q_j - \frac{1}{2} q_i K_{ik}.$$

The final expression has two terms, but they are essentially the same. They are just implicit sums with different dummy indices i and j . If we use the symmetric spring constant matrix $K_{ij} = K_{ji}$,

$$\frac{\partial \mathcal{L}}{\partial q_k} = -\frac{1}{2} K_{kj} q_j - \frac{1}{2} K_{ki} q_i = -K_{ki} q_i.$$

Similarly,

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_k} = M_{ki} \dot{q}_i \implies \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_k} = M_{ki} \ddot{q}_i.$$

By Lagrange equations, the equations of motion are

$$M_{ji} \ddot{q}_i = -K_{ji} q_i \quad \text{or} \quad \hat{\mathbf{M}} \ddot{\mathbf{q}} = -\hat{\mathbf{K}} \mathbf{q}. \quad (5.2)$$

Recall that in the simplest case, we decouple the equations by making a change of coordinates $\ddot{\xi}_i = -\omega_i^2 \xi_i$ (with no sum over i). We cannot inspect and guess what the coordinate transformation is, so assume that the new coordinates are

$$\xi_i = P_{ij}^{-1} q_j \quad \text{or} \quad \boldsymbol{\xi} = \hat{\mathbf{P}}^{-1} \mathbf{q},$$

where P_{ij}^{-1} are some linear transformation of the old coordinates. Substituting the coordinate transformation into the equations of motion (5.2),

$$\hat{\mathbf{M}} \hat{\mathbf{P}} \ddot{\boldsymbol{\xi}} = -\hat{\mathbf{K}} \hat{\mathbf{P}} \boldsymbol{\xi} \quad \implies \quad \ddot{\boldsymbol{\xi}} = -\hat{\mathbf{P}}^{-1} \hat{\mathbf{M}}^{-1} \hat{\mathbf{K}} \hat{\mathbf{P}} \boldsymbol{\xi}.$$

In index notation, this is $\ddot{\xi}_i = -P_{ij}^{-1} (M^{-1} K)_{ik} P_{kl} \xi_l$. Now $\ddot{\boldsymbol{\xi}}$ is just some complicated matrix times the transformed coordinates $\boldsymbol{\xi}$. For this system of equations to be decoupled, $\hat{\mathbf{P}}^{-1} \hat{\mathbf{M}}^{-1} \hat{\mathbf{K}} \hat{\mathbf{P}}$ must be a diagonal matrix. Luckily, from linear algebra, we know that this matrix is diagonalized if the columns of $\hat{\mathbf{P}}$ are eigenvectors of $\hat{\mathbf{M}}^{-1} \hat{\mathbf{K}}$. The diagonal elements of $\hat{\mathbf{P}}^{-1} \hat{\mathbf{M}}^{-1} \hat{\mathbf{K}} \hat{\mathbf{P}}$ are the corresponding eigenvalues of $\hat{\mathbf{M}}^{-1} \hat{\mathbf{K}}$. Physically, the eigenvalues are the ω_i^2 's.

Example 5.1. Two masses three springs again

Back to the simplest case, the equations of motion are

$$\begin{cases} m\ddot{x}_1 = -kx_1 - k'(x_1 - x_2), \\ m\ddot{x}_2 = -kx_2 + k'(x_1 - x_2). \end{cases}$$

We recognize the mass matrix and spring constant matrix to be

$$\hat{\mathbf{M}} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}, \quad \hat{\mathbf{K}} = \begin{bmatrix} k+k' & -k' \\ -k' & k+k' \end{bmatrix} \quad \implies \quad \hat{\mathbf{M}}^{-1} \hat{\mathbf{K}} = \begin{bmatrix} \frac{k+k'}{m} & -\frac{k'}{m} \\ -\frac{k'}{m} & \frac{k+k'}{m} \end{bmatrix}.$$

Now this is just an eigenvalue problem. We can solve it by hand or leave it to a computer. The eigenvalues are ω^2 , so

$$0 = \begin{vmatrix} \frac{k+k'}{m} - \omega^2 & -\frac{k'}{m} \\ -\frac{k'}{m} & \frac{k+k'}{m} - \omega^2 \end{vmatrix} = \left(\frac{k+k'}{m} - \omega^2 \right) - \frac{k'^2}{m^2}.$$

This equation can be solved by the quadratic formula, giving

$$\omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{k+2k'}{m}.$$

These are exactly the normal mode frequencies. Their corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1/2 \\ 1/2 \end{bmatrix}.$$

The matrix $\hat{\mathbf{P}}$ are composed of the eigenvectors, and the transformation matrix is $\hat{\mathbf{P}}^{-1}$:

$$\hat{\mathbf{P}} = \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}, \quad \hat{\mathbf{P}}^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

As a check,

$$\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 - x_1 \end{bmatrix}.$$

Usually, after finding the normal modes, the job is done because the coordinates ξ_i are the “natural” way to describe oscillations,

$$\xi_i = A_i \cos(\omega_i t + \phi_i),$$

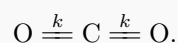
for A_i and ϕ_i fixed by initial conditions. The normal mode frequencies are the “natural frequencies”. However, one can still be interested in the q_i coordinates. By direct substitution,

$$q_i = P_{ij} \xi_j = \sum_j A_j v_{j,i} \cos(\omega_j t + \phi_j),$$

where $v_{j,i}$ is the i th component of the j th eigenvector of $\hat{\mathbf{M}}^{-1} \hat{\mathbf{K}}$, and ω_j^2 is the j th eigenvalue of it.

Example 5.2. Carbon Dioxide Molecule

Consider a carbon dioxide (CO_2) molecule:



It can be seen as a system of coupled oscillators with three masses and two springs. Let the two oxygen atoms have mass M and the carbon atoms have mass m . Each oxygen atom is connected to the central carbon atom with a spring of spring constant k . The three atoms are colinear. Let x_1, x_2, x_3 denotes the displacement of O, C, O respectively. The Lagrangian is given by

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} M (\dot{x}_1^2 + \dot{x}_3^2) + \frac{1}{2} m \dot{x}_2^2 - \frac{1}{2} k (x_1 - x_2)^2 - \frac{1}{2} k (x_2 - x_3)^2 \\ &= \frac{1}{2} M \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2 + \frac{1}{2} M \dot{x}_3^2 - \frac{1}{2} k x_1^2 - k x_1 x_2 - k x_2^2 - k x_2 x_3 - \frac{1}{2} k x_3^2. \end{aligned}$$

The mass and spring matrices are

$$M_{ij} = \begin{bmatrix} m & & \\ & m & \\ & & m \end{bmatrix}_{ij}, \quad K_{ij} = \begin{bmatrix} 2k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & 2k \end{bmatrix}_{ij}.$$

The eigenvalues and eigenvectors are

$$\omega_1^2 = 0, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}; \quad \omega_2^2 = \frac{k}{M}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}; \quad \omega_3^2 = \left(1 + \frac{2M}{m}\right) \frac{k}{M}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ -2M/m \\ 1 \end{bmatrix}.$$

From the eigenvectors and eigenvalues, we can know how the three normal modes behave. The first case has all three atoms oscillating in one direction, with frequency zero (or period infinity). This is just saying that there are no relative motion between all atoms, and the molecule is moving at a constant velocity. We now know that translational motion can also be a normal mode. The second case has two oxygen atoms moving in opposite direction with the same amplitude and frequency, while the central carbon stays at rest. The third case has the carbon atom oscillating in opposite direction relative to the two oxygen atoms and with amplitude $2M/m$ greater than oxygen atoms.

If we allow this CO_2 to move in three dimensions. There are six more degrees of freedoms, and hence six additional normal modes. Among these six additional ones, two are translational (along y and z), two are rotational (about y - and z -axis), and two are vibrational with atoms oscillating on along a straight line.

5.4 Waves

5.4.1 The Wave Equation

Consider the continuum limit of a system of coupled oscillators, in which there are infinite number of infinitesimal masses, each connected to its neighbors by infinitesimal springs. This is a typical model of a solid in one dimension. Let each mass have mass m , and each spring have spring constant k and equilibrium separation a . The Lagrangian for the system is

$$\mathcal{L} = \sum_i \left[\frac{1}{2} m \dot{x}_i^2 - \frac{1}{2} k (x_{i+1} - x_i)^2 \right],$$

where x_i is the displacement from equilibrium position of the i th mass. When we are taking the continuum limit such as a solid, it must have some finite length and mass. This means the separation must have $a \rightarrow 0$ and $m_i \rightarrow 0$. For the whole system to stretch by some amount, each spring must also stretch a small amount from the equilibrium. These springs are connected in series, so it requires $k \rightarrow \infty$ so the spring constant of the whole system is nonzero.

Instead of using index i in the continuum limit, we should turn to a continuous function $\eta(t, x)$ to describe the system. The η function is the instantaneous displacement of the mass from its equilibrium position x . Then we can write

$$\dot{x}_i(t) \rightarrow \frac{\partial \eta}{\partial t}, \quad x_{i+1} - x_i \rightarrow \eta(x+a) - \eta(x) \approx a \frac{\partial \eta}{\partial x}.$$

The action is

$$S \approx \int dt \sum_{\text{rod}} \left[\frac{1}{2} m \left(\frac{\partial \eta}{\partial t} \right)^2 - \frac{1}{2} k a^2 \left(\frac{\partial \eta}{\partial x} \right)^2 \right] \rightarrow \int dt \int dx \left[\frac{1}{2} \frac{m}{a} \left(\frac{\partial \eta}{\partial t} \right)^2 - \frac{1}{2} k a \left(\frac{\partial \eta}{\partial x} \right)^2 \right].$$

In the last step, we use the limit $a = dx$ to change the sum into an integral. The quantity $\mu \equiv m/a$ is the mass density and $Y \equiv ka$ is called **Young's modulus**. The action becomes

$$S[\eta] = \int dt dx \left[\frac{1}{2} \mu \left(\frac{\partial \eta}{\partial t} \right)^2 - \frac{1}{2} Y \left(\frac{\partial \eta}{\partial x} \right)^2 \right] = \int dt dx \mathcal{L}.$$

The quantity in the square bracket is now the **Lagrangian density**. Integrating it over all space gives the Lagrangian. The Lagrangian density is a function of two independent variables (t, x) . To extremize action, the Euler-Lagrange equations generalize to

$$0 = \frac{\partial \mathcal{L}}{\partial \eta} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\eta}} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial \eta'}, \quad \text{where} \quad \dot{\eta} = \frac{\partial \eta}{\partial t}, \quad \eta' = \frac{\partial \eta}{\partial x}.$$

Applying the Euler-Lagrange equation to the Lagrangian density, we obtain the **wave equation**:

$$\frac{\partial^2 \eta}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 \eta}{\partial t^2} = 0, \tag{5.3}$$

where $v \equiv \sqrt{Y/\mu}$ is the speed of the wave. Since the masses are in one dimension, this wave is a longitudinal wave. In fact, the same result holds for transverse wave when we generalize to higher dimensions. It is easy to check that the solution to $\eta(t, x)$ is of the form

$$\eta(t, x) = A f(x + vt) + B g(x - vt),$$

where f and g are arbitrary functions, and A and B are constants. The functions f and g specifies the shape of the wave. Meanwhile, $f(x + vt)$ is the piece that travels to the left (or $-x$) and $g(x - vt)$ travels to the right (or $+x$).

The most useful shape functions are $e^{ik(x \pm vt)}$. Physically, $e^{ik(x \pm vt)}$ are plane waves. These solutions are analogous to normal modes: complex exponentials are periodic, so $\eta \propto Ae^{ik(x \pm vt)}$ has every mass in the solid oscillating at the same frequency kv . Suppose the object has length L , and we let η to be periodic in L (i.e. $\eta(0) = \eta(L)$). The possible values of k are $2\pi n/L$, so the normal frequencies are $2\pi nv/L$, where $n = 0, 1, 2, 3, \dots$.

5.4.2 Fourier Series and Fourier Transform

Mathematically, we can express any shape with periodic boundary condition $\eta(0) = \eta(L)$ in terms of plane wave solutions:

$$\eta(t, x) = \sum_n \left[A_n e^{ik_n(x-vt)} + B_n e^{ik_n(x+vt)} \right].$$

The coefficients A_n and B_n can be fixed by initial conditions, namely

$$\eta(0, x) = \sum_n (A_n e^{ik_n x} + B_n e^{ik_n x}) = \sum_n (A_n + B_n) e^{ik_n x},$$

$$\dot{\eta}(0, x) = \sum_n (-ik_n v A_n e^{ik_n x} + ik_n v B_n e^{ik_n x}) = \sum_n ik_n v (B_n - A_n) e^{ik_n x}.$$

If we know the length L and the wavenumbers $k_n = 2\pi n/L$, we can obtain these coefficients by invoking orthogonality of $e^{ik_n x}$,

$$\int_0^L dx e^{i(k_n - k_m)x} = L \delta_{mn}.$$

Thus,

$$\int_0^L dx \eta(0, x) e^{-ik_m x} = \int_0^L dx \sum_n (A_n + B_n) e^{ik_n x} e^{-ik_m x} = \sum_n (A_n + B_n) \int_0^L dx e^{i(k_n - k_m)x} = L(A_m + B_m).$$

We can use the same trick for $\dot{\eta}(0, x)$ and finally obtain

$$A_n + B_n = \frac{1}{L} \int_0^L dx \eta(0, x) e^{-ik_n x}, \quad ik_n v (B_n - A_n) = \frac{1}{L} \int_0^L dx \dot{\eta}(0, x) e^{-ik_n x}.$$

The individual coefficients are

$$A_n = \frac{1}{2L} \int_0^L dx \left[\eta(0, x) + \frac{iL}{2\pi nv} \dot{\eta}(0, x) \right] e^{-ik_n x}, \quad B_n = \frac{1}{2L} \int_0^L dx \left[\eta(0, x) - \frac{iL}{2\pi nv} \dot{\eta}(0, x) \right] e^{-ik_n x}.$$

Alternatively, can rewrite the Fourier series as

$$\eta(x) = \sum_{n=-\infty}^{\infty} \tilde{\eta}(n) e^{2\pi i n x / L} \quad \text{and} \quad \eta(n) = \frac{1}{L} \int_{-L/2}^{L/2} \eta(x) e^{-2\pi i n x / L} dx.$$

By extending $L \rightarrow \infty$, $k = 2\pi n/L$, $L\eta(n) = \tilde{\eta}(k)$, and assuming $\eta(\pm\infty) \rightarrow 0$ as boundary condition:

$$\eta(x) = \sum_{n=-\infty}^{\infty} \eta(n) e^{2\pi i n x / L} = \int_{-\infty}^{+\infty} \eta(n) e^{2\pi i n x / L} dn = \int_{-\infty}^{+\infty} \tilde{\eta}(k) e^{ikx} \left(\frac{L}{2\pi} \right) dk = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{\eta}(k) e^{ikx} dk.$$

The quantity $\tilde{\eta}(k)$ in k -space is called the **Fourier transform** of $\eta(x)$,

$$\tilde{\eta}(k) = \int_{-\infty}^{+\infty} \eta(x) e^{-ikx} dx,$$

and $\eta(x)$ is the **inverse Fourier transform** of $\tilde{\eta}(k)$,

$$\eta(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{\eta}(k) e^{ikx} dk.$$

There are many conventions (involving where should we put the factor 2π) to write Fourier transform and inverse Fourier transform. Another useful convention is

$$\tilde{\eta}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \eta(x) e^{-ikx} dx \quad \Longleftrightarrow \quad \eta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \tilde{\eta}(k) e^{ikx} dk.$$

6 HAMILTONIAN MECHANICS

Compared to Newtonian mechanics, the Lagrangian formulation together with the concept of action has many advantages. In particular, it works a lot easier in non-Cartesian coordinates; it is able to indicate various conserved quantities; it also handles non-inertial frames and coupled systems well. In this chapter, we will discuss an even more powerful tool than the Lagrangian formulation—the Hamiltonian formulation.

We know that in Lagrangian mechanics, the fundamental variables are in the **configuration space**, namely q and its derivative $\dot{q}$. The velocity term $\dot{q}$ is sometimes hard to work with. For example, when deriving Newton’s second law in polar coordinates, we encounter a term like $r\dot{\phi}$, which is the Coriolis force. It has a unit of acceleration, but is conceptually not something like a usual acceleration. Thus, in Hamiltonian mechanics, we will turn to the **phase space**, using coordinates q and momentum p instead of velocity $\dot{q}$. The reason is simple, in many cases we will have the conservation of momentum, but we have never heard of “the conservation of velocity”. The momentum should be a natural variable to work with.

6.1 Legendre Transformation

In Lagrangian mechanics, the generalized momentum is defined as

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}.$$

We want to transform the Lagrangian (a function of $q, \dot{q}$) to a new function of q, p . For transformation between variables especially when they are related in the above definition, it is most useful to consider a **Legendre transformation**. Legendre transformations are most useful in thermal dynamics and statistical mechanics. For example, the thermodynamic identity tells that

$$dU = T dS - P dV + \mu dN.$$

These are the variables: the entropy S , the volume V , and the particle number N . If dV and dN are both zero, we have the definition of temperature: $T = (\partial U / \partial S)_{V, N}$. In general, it is hard to control the entropy of a system. It is more convenient to control the temperature, so we want something like dT . Legendre transformation tells us that we can consider a new quantity $F = U - TS$ (called the Helmholtz free energy). Note that all we do is adding a $-TS$ to the energy U . Now compute a new thermodynamic identity, and see how the free variable changes from S to T :

$$dF = dU - T dS - S dT = T dS - P dV + \mu dN - T dS - S dT = -S dT - P dV + \mu dN.$$

Now T is the free variable. All we have used is the product rule, which is the core of Legendre transformations.

Anyways, all we need to know is that the Legendre transform of the Lagrangian is the Hamiltonian:

$$H = p_i \dot{q}_i - \mathcal{L}.$$

(Note the product $p_i \dot{q}_i$.) The natural variables of the Hamiltonian are q_i , p_i and t . All the $\dot{q}_i$ in the Lagrangian are replaced by

$p_i = \partial\mathcal{L}/\partial\dot{q}_i$. For instance, consider the one-dimensional harmonic oscillator

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2, \quad p_x = \frac{\partial\mathcal{L}}{\partial\dot{x}} = m\dot{x}.$$

The Hamiltonian is

$$H = p_x\dot{x} - \mathcal{L} = p_x\frac{p_x}{m} - \frac{1}{2}m\left(\frac{p_x}{m}\right)^2 + \frac{1}{2}kx^2 = \frac{p_x^2}{2m} + \frac{1}{2}kx^2.$$

6.2 Hamilton's equations

To find the equation of motions for the system from the Hamiltonian, just extremize the action. In terms of the Hamiltonian, the action is

$$S = \int_{t_1}^{t_2} \mathcal{L} dt = \int_{t_1}^{t_2} (p_i\dot{q}_i - H) dt.$$

The change in action under a small variation of trajectory is

$$\delta S = S[q_i + \delta q_i, p_i + \delta p_i] - S[q_i, p_i] = \int_{t_1}^{t_2} \left[\delta p_i \frac{dq_i}{dt} + p_i \frac{d(\delta q_i)}{dt} - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right] dt.$$

We need $\delta S = 0$ at the physical trajectory and $\delta q_i(t_1) = \delta q_i(t_2) = 0$ for all possible trajectories. Integrating the second term by parts,

$$\delta S = \int_{t_1}^{t_2} \left(\frac{dq_i}{dt} \delta p_i - \frac{dp_i}{dt} \delta q_i - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right) dt + \cancel{p_i \delta q_i \Big|_{t_1}^{t_2}} = \int_{t_1}^{t_2} \left[\left(-\frac{dp_i}{dt} - \frac{\partial H}{\partial q_i} \right) \delta q_i + \left(\frac{dq_i}{dt} - \frac{\partial H}{\partial p_i} \right) \delta p_i \right] dt.$$

The last term in the first equality vanishes because $\delta q_i(t_1) = \delta q_i(t_2) = 0$. To have $\delta S = 0$ for arbitrary variations, the coefficients of δq_i and δp_i must also vanish. This gives two **Hamilton's equations** for each i :

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}} \quad (6.1)$$

Hamilton's equations are the equations of motion that tells how q_i and p_i evolve in time. Instead of having N second-order differential equations for N coordinates q_i in Lagrangian mechanics, we have $2N$ first-order differential equations for the $2N$ phase space coordinates q_i, p_i . One immediate conclusion drawn from Hamilton's equations is that

$$\frac{dH}{dt} = \frac{\partial H}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial H}{\partial p_i} \frac{dp_i}{dt} + \frac{\partial H}{\partial t} = \frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} + \frac{\partial H}{\partial t} = \frac{\partial H}{\partial t}.$$

The Hamiltonian is conserved along physical trajectories iff it does not explicitly depend on time.

In phase space, q_i and p_i are *independent* variables and they act like coordinates. This is why we are able to make coordinate transformation in phase space. Physical trajectories trace out a curve in phase space. This cannot be done if we use q_i and $\dot{q}_i$ as in Lagrangian mechanics because $\dot{q}_i$ depends strongly on q_i . Obviously, they are not independent variables.

6.3 Coordinate Transformations

6.3.1 Unified Coordinates and Canonical Transformation

Once we are in phase space, it does not matter where the coordinate axes are. Just like ordinary spaces, a vector is the same vector after you make a coordinate transformation. Manipulating the axes does nothing but changing the components of the vector. Thus, it is possible to mix q_i and p_i into some (not necessarily linear) combinations that makes solving problem easy. Now look again at Hamilton's equations,

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

The two equations look very similar if q_i and p_i are treated the same. It is convenient to define **unified coordinates** ξ_α , where α (and all other Greek letters) runs from 1 to $2N$ for a system with N degrees of freedom.

$$\xi_1 = q_1, \xi_2 = q_2, \dots, \xi_N = q_N, \quad \xi_{N+1} = p_1, \xi_{N+2} = p_2, \dots, \xi_{2N} = p_N.$$

Hamilton's equations can then be written in a compact form:

$$\dot{\xi}_\alpha = \omega_{\alpha\beta} \frac{\partial H}{\partial \xi_\beta}, \quad \omega_{\alpha\beta} = \begin{bmatrix} O_{N \times N} & I_{N \times N} \\ -I_{N \times N} & O_{N \times N} \end{bmatrix}_{\alpha\beta}. \quad (6.2)$$

The matrix $\omega_{\alpha\beta}$ is known as the **symplectic matrix**. Consider a coordinate transformation of ξ_β to some new coordinates η_α . By the chain rule,

$$\dot{\eta}_\alpha = \frac{\partial \eta_\alpha}{\partial \xi_\beta} \frac{d\xi_\beta}{dt} = \frac{\partial \eta_\alpha}{\partial \xi_\beta} \omega_{\beta\gamma} \frac{\partial H}{\partial \xi_\gamma} = \left(\omega_{\beta\gamma} \frac{\partial \eta_\alpha}{\partial \xi_\beta} \frac{\partial \eta_\delta}{\partial \xi_\gamma} \right) \frac{\partial H}{\partial \eta_\delta} \stackrel{!}{=} \omega_{\alpha\delta} \frac{\partial H}{\partial \eta_\delta}.$$

In the second equality, we used Hamilton's equations. As desired, the final form of this time derivative is some coefficients times $\partial H / \partial \eta_\delta$. This is Hamilton's equations in the new coordinate. However, these coefficients seem very ugly, which might make Hamilton's equations hard to work with. In order to solve this problem, we will only make coordinate transformations such that Hamilton's equations look the same in both coordinates. In other words, the **symplectic condition** should be satisfied:

$$\omega_{\alpha\delta} = \omega_{\beta\gamma} \frac{\partial \eta_\alpha}{\partial \xi_\beta} \frac{\partial \eta_\delta}{\partial \xi_\gamma}.$$

Such a coordinate transformation is called the **canonical transformation**. The coordinates given by a coordinate transformation are **canonical coordinates**.

This should remind us of Lorentz transformations in special relativity. Lorentz transformations ensure that the Minkowski metric is invariant in new coordinates:

$$\eta_{\mu'\nu'} = \eta_{\mu\nu} \frac{\partial r^\mu}{\partial r^{\mu'}} \frac{\partial r^\nu}{\partial r^{\nu'}}.$$

It turns out that this works in all inertial frames. It is definitely possible to work in non-inertial frames and find how the metric changes. (In fact, this is what general relativity does.) Similarly, it is allowed to make “non-canonical” transformations, but we then need to figure out Hamilton's equations by ourselves. It is more easy to work in canonical coordinates.

6.3.2 Poisson Brackets

The symplectic condition is so useful and important that the term on the right side gets a name known as **Poisson brackets**:

$$\{A, B\} \equiv \omega_{\alpha\beta} \frac{\partial A}{\partial \xi_\alpha} \frac{\partial B}{\partial \xi_\beta} = \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i}. \quad (6.3)$$

Here are some properties of the Poisson bracket:

- Anti-commutativity: $\{A, B\} = -\{B, A\}$.
- Distributivity: $\{A, bB + cC\} = b\{A, B\} + c\{A, C\}$.
- Modified associativity: $\{AB, C\} = \{A, C\}B + A\{B, C\}$.
- Satisfies the Jacobi identity: $\{A, \{B, C\}\} + \{B, \{C, A\}\} + \{C, \{A, B\}\} = 0$.

We want

$$\omega_{\beta\gamma} \frac{\partial \eta_\alpha}{\partial \xi_\beta} \frac{\partial \eta_\delta}{\partial \xi_\gamma} = \omega_{\alpha\delta} \implies \{\eta_\alpha, \eta_\delta\}_\xi = \omega_{\alpha\delta}.$$

Define new position and momentum coordinates $Q_i = \eta_i$ and $P_i = \eta_{N+i}$. This means

$$\{Q_i, Q_j\} = 0, \quad \{P_i, P_j\} = 0, \quad \{Q_i, P_j\} = \delta_{ij}. \quad (6.4)$$

Poisson brackets are anti-symmetric, so we don't need to write a separation condition for $\{P_i, Q_j\}$. It turns out that the Poisson bracket is independent of canonical coordinates:

$$\{A, B\}_\xi = \omega_{\alpha\beta} \frac{\partial A}{\partial \xi_\alpha} \frac{\partial B}{\partial \xi_\beta} = \omega_{\alpha\beta} \frac{\partial A}{\partial \eta_\gamma} \frac{\partial \eta_\gamma}{\partial \xi_\alpha} \frac{\partial B}{\partial \eta_\delta} \frac{\partial \eta_\delta}{\partial \xi_\beta} = \left(\omega_{\alpha\beta} \frac{\partial \eta_\gamma}{\partial \xi_\alpha} \frac{\partial \eta_\delta}{\partial \xi_\beta} \right) \frac{\partial A}{\partial \eta_\gamma} \frac{\partial B}{\partial \eta_\delta} = \omega_{\gamma\delta} \frac{\partial A}{\partial \eta_\gamma} \frac{\partial B}{\partial \eta_\delta} = \{A, B\}_\eta.$$

This is why we don't need to write subscripts for Poisson brackets.

To see how to work with Poisson brackets and canonical transformations, we will do some examples. The first system a physicist should think of at any time is definitely the harmonic oscillator.

Example 6.1. The Harmonic Oscillator

The harmonic oscillator has a simple Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2.$$

It can be seen as a square p^2 plus another square q^2 , each with some coefficients. This is like an ellipse in a two-dimensional phase space, which suggests some coordinate transformation to “polar coordinates”. Let

$$q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2m\omega P} \cos Q.$$

(One can also try $q \sim \cos Q$ and $p \sim \sin Q$, but will find that the Poisson bracket acquires an additional minus sign.) The Hamiltonian simplifies to

$$H = \frac{1}{2m}(2m\omega P) \cos^2 Q + \frac{1}{2}m\omega^2 \frac{2P}{m\omega} \sin^2 Q = \omega P.$$

To see whether this is a canonical transformation, just compute the Poisson brackets:

$$\{q, p\} = \frac{\partial q}{\partial Q} \frac{\partial p}{\partial P} - \frac{\partial q}{\partial P} \frac{\partial p}{\partial Q} = \left(\sqrt{\frac{2P}{m\omega}} \cos Q \right) \left(\sqrt{\frac{m\omega}{2P}} \cos Q \right) - \left(\sqrt{\frac{1}{2Pm\omega}} \sin Q \right) \left(-\sqrt{2m\omega P} \sin Q \right) = 1.$$

Because partial derivatives commute, $\{q, q\}$ and $\{p, p\}$ are trivially zero. The two Hamilton's equations are extremely simple

$$\dot{Q} = \frac{\partial H}{\partial P} = \omega, \quad \dot{P} = -\frac{\partial H}{\partial Q} = 0.$$

The solutions are

$$Q = \omega t + Q_0, \quad P = P_0.$$

Substituting them back to original coordinates:

$$q = \sqrt{\frac{2P_0}{m\omega}} \cos(\omega t + Q_0), \quad p = \sqrt{2m\omega P_0} \sin(\omega t + Q_0).$$

Both Q and P have physical meanings and they may tell us something useful. First, P is the energy divided by frequency (it has units of angular momentum), while Q is the usual phase in solutions of simple harmonic oscillator. So those coordinates describe the energy and the phase of the trajectory. Since $P = P_0$, we directly know that the energy of the system is constant, as expected.

6.4 Conserved Quantities and Noether's Theorem

6.4.1 The Generator Function

In Example 6.1, we proposed a canonical transformation. In many cases, it is often hard to come up with a transformation by inspection, so we need some methods that help us find it. The equations of motion comes from extremizing the action:

$$S = \int_{t_1}^{t_2} (p_i \dot{q}_i - H) dt.$$

If a coordinate transformation also gives equations of motion in the form of Hamilton's equations, the action should also look like

$$S = \int_{t_1}^{t_2} (P_i \dot{Q}_i - H) dt.$$

For arbitrary transformations, the Hamiltonian is always the Hamiltonian, but $p_i \dot{q}_i$ may not be $P_i \dot{Q}_i$. This is one disadvantage of Hamiltonian mechanics: the action depends explicitly on coordinates or their derivatives. In Lagrangian mechanics there is no such a problem because the Lagrangian is coordinate invariant. What we will do now is to relax the integrand a bit. That is, we do not require the integrand to be the same, but back to the bottom line: the equations of motion should look the same.

First, adding a total time-derivative to the integrand does nothing to equations of motion:

$$\int_{t_1}^{t_2} \frac{dF}{dt} dt = F(t_2) - F(t_1).$$

Since the endpoints are fixed, this has no effect on varying the trajectories and evaluating δS . This is a similar argument to Theorem 2.1, which says that adding a total time-derivative to the Lagrangian produce the same equations of motion. One good thing about Hamiltonian mechanics is that we do not require the Hamiltonian to be “the Hamiltonian”. What does this mean? Consider the following action,

$$S = \int_{t_1}^{t_2} \left(P_i \dot{Q}_i - K + \frac{dF}{dt} \right) dt.$$

It produces some sort of Hamilton's equations, but with H replaced with some function K .

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i}.$$

The fake Hamiltonian has a name, often called the “Kamiltonian”. Assume that $F = F(q, P, t)$. By the chain rule,

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial q_k} \dot{q}_k + \frac{\partial F}{\partial P_k} \dot{P}_k.$$

Then we arrive at

$$P_i \dot{Q}_i - K + \frac{\partial F}{\partial t} + \frac{\partial F}{\partial q_k} \dot{q}_k + \frac{\partial F}{\partial P_k} \dot{P}_k \stackrel{?}{=} p_i \dot{q}_i - H.$$

If both sides were equal, we should choose the following:

$$p_i = \frac{\partial F}{\partial q_i}, \quad K = H + \frac{\partial F}{\partial t}, \quad Q_i = \frac{\partial F}{\partial P_i}.$$

Substitute these into the equation,

$$P_i \dot{Q}_i - H - \frac{\partial F}{\partial t} + \frac{\partial F}{\partial t} + p_k \dot{q}_k + Q_k \dot{P}_k \stackrel{?}{=} p_i \dot{q}_i - H,$$

leaving

$$0 \stackrel{?}{=} P_i \dot{Q}_i + Q_i \dot{P}_i = \frac{d}{dt}(P_i Q_i).$$

The RHS is in general not zero, but a total time-derivative is acceptable in the action integrand. In conclusion, given some function $F(q, P, t)$, we can solve for Q_i and P_i as functions of q_i and p_i using

$$p_i = \frac{\partial F}{\partial q_i}, \quad Q_i = \frac{\partial F}{\partial P_i}, \quad \text{and} \quad K = H + \frac{\partial F}{\partial t}. \quad (6.5)$$

The function F is known as the **generator** of the corresponding canonical transformation.

Example 6.2. The identity transformation

Consider the generator $F(q, P, t) = q_i P_i$. Then the canonical transformation is

$$p_i = \frac{\partial F}{\partial q_i} = P_i, \quad Q_i = \frac{\partial F}{\partial P_i} = q_i, \quad \text{and} \quad K = H + \frac{\partial F}{\partial t} = H.$$

Everything is unchanged, so this transformation is just the identity. Since the original coordinates q_i, p_i satisfy Hamilton's equations, the identity transformation must be a canonical transformation.

Example 6.3. The point transformation

Consider the generator $F(q, P, t) = f_i(q) P_i$, where f_i are a set of N functions, each of which can depend on all of the q . This gives

$$Q_i = f_i(q), \quad p_i = \frac{\partial F}{\partial q_i} = \frac{\partial f_k}{\partial q_i} P_k, \quad \text{and} \quad K = H + \frac{\partial F}{\partial t} = H.$$

Looking at $Q_i = f_i(q)$, we see that f_i tells how coordinates q transform to Q . The momentum relations are a bit subtle. If we were in Lagrangian mechanics and were seeking generalized momenta corresponding to the new generalized coordinates,

$$P'_i = \frac{\partial \mathcal{L}}{\partial \dot{Q}_i},$$

The old momentum coordinates can be expressed as

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{\partial \mathcal{L}}{\partial \dot{Q}_k} \frac{\partial \dot{Q}_k}{\partial \dot{q}_i} = \frac{\partial \mathcal{L}}{\partial \dot{Q}_k} \frac{\partial Q_k}{\partial q_i} = P'_k \frac{\partial f_k}{\partial q_i}.$$

This is exactly what we have from the generator. The new momentum coordinates are the usual generalized momenta for the new spatial coordinates in Lagrangian mechanics. Inverting the matrix of partial derivatives $\partial f_k/\partial q_i$ gives the expressions for P in terms of p . These transformations are called **point transformations**. Point transformations are the full set of coordinate transformations in Lagrangian mechanics, but they are only a subset of canonical transformations in Hamiltonian mechanics.

6.4.2 Infinitesimal Transformation

Consider the generator

$$F = q_i P_i + \epsilon G(q_i, P_i),$$

where G is some function of q_i and P_i and $\epsilon \ll 1$. From Example 6.2, we know that $F = q_i P_i$ generates the identity transformation, so this F will generate something that is almost the identity transformation. Using 6.5, we can compute

$$Q_i = q_i + \epsilon \frac{\partial G}{\partial P_i}, \quad p_i = P_i + \epsilon \frac{\partial G}{\partial q_i} \implies P_i = p_i - \epsilon \frac{\partial G}{\partial q_i}.$$

By the chain rule, the Q_i equation is

$$Q_i = q_i + \epsilon \frac{\partial G}{\partial p_i} \frac{\partial p_i}{\partial P_i} = q_i + \epsilon \frac{\partial G}{\partial p_i} + \mathcal{O}(\epsilon^2).$$

The second order terms are negligible. These statements can be rewritten in terms of unified coordinates and Poisson brackets:

$$\eta_\alpha = \xi_\alpha + \epsilon \{\xi_\alpha, G\}.$$

To check this, first calculate $\{\xi_\alpha, G\}$,

$$\{\xi_\alpha, G\} = \omega_{\beta\gamma} \frac{\partial \xi_\alpha}{\partial \xi_\beta} \frac{\partial G}{\partial \xi_\gamma} = \omega_{\beta\gamma} \delta_{\alpha\beta} \frac{\partial G}{\partial \xi_\gamma} = \omega_{\alpha\gamma} \frac{\partial G}{\partial \xi_\gamma}.$$

If α is between 1 to N ($\xi_\alpha = q_i$ and $\eta_\alpha = Q_i$), then the only nonzero entry in α th row of $\omega_{\alpha\gamma}$ is at $\gamma = (\alpha + N)$ th column, which is $+1$ and corresponds to $\xi_\gamma = p_i$. This gives $Q_i = q_i + \epsilon(\partial G/\partial p_i)$. Similarly, if α is between $N + 1$ to $2N$ ($\xi_\alpha = p_i$ and $\eta_\alpha = P_i$), then the only nonzero entry in α th row of $\omega_{\alpha\gamma}$ is $\gamma = (\alpha - N)$ th column, which is -1 and corresponds to $\xi_\gamma = q_i$. This gives $P_i = p_i - \epsilon(\partial G/\partial q_i)$. This is useful, because given some $G(q_i, P_i)$, we can find a canonical transformation by shifting all unified coordinates by the Poisson bracket of the coordinate with G . The quantity G is also known as the **generator**. Moreover, the Poisson bracket with G can tell how *any* quantity transforms,

$$\delta A = A(\eta_\alpha) - A(\xi_\alpha) = A(\xi_\alpha + \epsilon \{\xi_\alpha, G\}) - A(\xi_\alpha) = (\epsilon \{\xi_\alpha, G\}) \frac{\partial A}{\partial \xi_\alpha} = \epsilon \omega_{\alpha\beta} \frac{\partial G}{\partial \xi_\beta} \frac{\partial A}{\partial \xi_\alpha} = \epsilon \{A, G\}. \quad (6.6)$$

Now consider the special case $G = H$, and $\eta_\alpha = \xi_\alpha + \epsilon \{\xi_\alpha, H\}$. The equations of motion is

$$\dot{\xi}_\alpha = \omega_{\alpha\beta} \frac{\partial H}{\partial \xi_\beta} = \{\xi_\alpha, H\} \implies \eta_\alpha = \xi_\alpha + \epsilon \{\xi_\alpha, H\} = \xi_\alpha + \epsilon \frac{d\xi_\alpha}{dt}.$$

In this case, the new coordinates η at time t is the old coordinates ξ at time $t + \epsilon$. Equivalently, it means that η represents the future of ξ . Since this is a canonical transformation, we can conclude that time evolution is a canonical transformation generated by the Hamiltonian. This again applies to other quantities that are not coordinates:

$$\frac{dA}{dt} = \frac{\partial A}{\partial \xi_\alpha} \frac{d\xi_\alpha}{dt} + \frac{\partial A}{\partial t} = \frac{\partial A}{\partial \xi_\alpha} \omega_{\alpha\beta} \frac{\partial H}{\partial \xi_\beta} + \frac{\partial A}{\partial t} = \{A, H\} + \frac{\partial A}{\partial t}. \quad (6.7)$$

In general, we assume (and it is often the case) that A has no explicit time dependence. Suppose we choose G to be some conserved quantity. Then

$$0 = \frac{dG}{dt} = \{G, H\} \iff \delta H = 0.$$

This means the Hamiltonian does not change under the canonical transformation generated by G . That is, this transformation is a symmetry of the Hamiltonian. The converse is also true: if a transformation is a symmetry of the Hamiltonian, then G is a conserved quantity. This is **Noether's theorem** in the Hamiltonian formulation.

Example 6.4. Translational Symmetry

Consider the momentum p_1 as the generator. The change in the Hamiltonian is

$$\delta H = \{H, G\} = \frac{\partial H}{\partial q_1} \frac{\partial p_1}{\partial p_1} = \frac{\partial H}{\partial q_1}.$$

This vanishes if H is independent of q_1 , which implies that if the Hamiltonian does not change under translations in q_1 , the corresponding momentum p_1 is conserved. It has the same result as in the Lagrangian formulation. Note that we do not specify which “momentum” p_1 is. It can be the usual linear momentum, or angular momentum (translation in q_1 is then the rotation in angle), or any other generalized momentum.

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